Journal of Open Innovation: Technology, Market, and Complexity 11 (2025) 100557

Contents lists available at ScienceDirect

Journal of Open Innovation: Technology, Market, and Complexity

journal homepage: www.sciencedirect.com/journal/journal-of-open-innovation-technology-market-and-complexity

Developing a hybrid machine learning model for employee turnover prediction: Integrating LightGBM and genetic algorithms

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ARTICLE INFO

Keywords:
Employee Turnover
Machine Learning
LightGBM
Genetic Algorithms
Feature Selection

ABSTRACT

Employee turnover poses a significant challenge to organizations, leading to increased costs and disruptions in workforce stability. In this study, we propose a novel hybrid model for employee turnover prediction that integrates Genetic Algorithms (GA) for feature selection with LightGBM for classification. The GA component identifies the most relevant features, while LightGBM ensures high prediction accuracy and computational efficiency. The model was evaluated on two datasets—a public HR Analytics dataset and a proprietary organizational dataset—using multiple performance metrics including Accuracy, Precision, Recall, F1-Score, and AUC. The proposed GA + LightGBM model achieved an AUC of 0.94 and F1-Score of 0.87 on the HR dataset, and an AUC of 0.78 with F1-Score of 0.73 on the organizational dataset, outperforming conventional models such as Random Forest, Logistic Regression, and even standalone LightGBM. The results demonstrate that the proposed method offers a balanced and interpretable solution for turnover prediction, outperforming existing approaches in both predictive power and practical applicability. This work contributes to HR analytics by offering a transparent, scalable, and high-performing predictive framework that supports effective retention strategies.

1. Introduction

Employee turnover remains one of the most significant challenges in human resource management, leading to increased costs related to recruitment, training, and the loss of valuable expertise (PM and Balaji, 2019). High turnover rates can disrupt organizational stability, reduce workforce morale, and negatively impact productivity. The ability to predict turnover accurately is essential for organizations to mitigate these costs and implement effective retention strategies. Traditional methods for predicting employee attrition, such as simple surveys or performance evaluations, often fail to capture the complexity of turnover decisions, as they overlook the intricate interplay of various factors affecting an employee's decision to leave the organization. (Marquez et al., 2024; Strohmeier & Piazza, 2015; H. Zhang et al., 2018). These conventional methods are too simplistic and cannot effectively account for the multi-dimensional nature of turnover that arises from both internal and external factors influencing employee decisions. In recent years, machine learning (ML) has emerged as a promising solution to improve turnover prediction (Al Akasheh et al., 2024; Atef et al., 2022; Park et al., 2024). Various machine learning algorithms, including decision trees, random forests, and support vector machines (SVM), have been employed to predict employee attrition by analyzing historical data, job satisfaction, performance metrics, and other relevant features (Alaskar et al., 2019; Veglio et al., 2025). For instance, Shafie et al. explored the integration of Artificial Neural Networks (ANN) with clustering techniques to enhance turnover prediction accuracy (Shafie et al., 2024), while Biswas et al. applied ensemble learning models to identify key predictors of turnover intention, showing improved performance compared to traditional methods (Biswas et al., 2023). Despite these advancements, many studies still struggle with handling high-dimensional datasets and the challenge of selecting the most relevant features that significantly influence turnover. (de Oliveira et al., 2019; Jain and Nayyar, 2018). To address these limitations, this paper proposes a novel approach that combines LightGBM, a state-of-the-art gradient boosting algorithm, with genetic algorithms for feature selection. LightGBM has gained popularity due to its ability to handle large-scale datasets efficiently, while genetic algorithms are used to optimize feature selection, ensuring that only the most impactful variables are included in the model. This hybrid approach aims to improve the accuracy and interpretability of employee turnover prediction models by leveraging the strengths of both algorithms. The proposed model is evaluated using two datasets: one publicly available,

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https://doi.org/10.1016/j.joitmc.2025.100557
Received 24 March 2025; Received in revised form 22 May 2025; Accepted 24 May 2025
Available online 27 May 2025
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HR_Analytics, and another collected by us through a questionnaire. The publicly available HR_Analytics dataset is widely used in employee attrition research, while the second dataset was specifically gathered for this study to provide additional insights into employee turnover. By including both publicly available and custom-collected data, this study ensures a broad representation of different organizational contexts and enhances the generalizability of the results. The results demonstrate significant improvements in prediction accuracy, with better performance in key metrics compared to traditional machine learning models.

This study offers three major contributions to the field of HR analytics and turnover prediction. First, it introduces a novel hybrid model that integrates Genetic Algorithms with LightGBM—a combination that has been rarely applied in previous research but shows strong potential for handling high-dimensional HR data. Second, it presents a unique dataset collected directly from real-world industrial organizations using a carefully designed questionnaire, thereby ensuring relevance and authenticity. Third, through comprehensive experimentation, it demonstrates the superior performance of the proposed hybrid approach over traditional models, validating its practical value for predicting employee attrition. These contributions collectively strengthen the foundation for data-driven HR decision-making and highlight the potential of intelligent algorithms in workforce analytics.

The structure of this paper is as follows: Section 2 provides a review of the related works and methodologies used in previous turnover prediction studies. Section 3 describes the proposed methodology, including the integration of LightGBM and genetic algorithms, and defines the key terms. Section 4 focuses on the selection and definition of other prediction methods used for comparison. Section 5 presents the statistical analysis of the data. Section 6 discusses the experimental results, comparing the performance of the proposed model with existing methods. Section 7 analyzes feature selection methods, comparing the computational complexity, training times, and performance of different feature selection techniques. Finally, Section 8 concludes the paper, summarizes the key findings, and outlines directions for future research. Figure 1 illustrates the overall roadmap of this research.

2. Literature review

Employee turnover prediction has been extensively studied in recent years, with various machine learning algorithms being applied to predict employee attrition (Ikram et al., 2021; Ilyas et al., 2020). This section reviews key works in the field, outlining the methodologies used and their findings, starting with the earliest studies and progressing through recent advancements.

Zhang et al. conducted a study to predict employee turnover using machine learning techniques (H. Zhang et al., 2018). They analyzed a dataset from a telecommunications company, focusing on employee characteristics such as job position, overtime, and work level. The study compared various machine learning algorithms like logistic regression, KNN, and decision trees to predict turnover. The results revealed that machine learning models could effectively capture the complex dynamics of employee attrition, with decision trees proving to be the most accurate in predicting turnover. The study emphasized the importance of identifying the most significant turnover predictors for targeted retention strategies.

Alaskar et al. explored the use of machine learning models, including decision trees, support vector machines, and logistic regression, for predicting employee turnover (Alaskar et al., 2019). They applied feature selection techniques, such as SelectKBest and Recursive Feature Elimination (RFE), to identify the most important variables impacting turnover decisions. The study showed that combining multiple machine learning models, such as SVM and random forests, enhanced the predictive accuracy of the model compared to traditional methods.

Zhao et al. expanded the understanding of turnover prediction by using a broader range of machine learning models to predict employee attrition (Zhao et al., 2019). Their study involved algorithms such as decision trees, random forests, logistic regression, and neural networks. The researchers found that including a variety of features, including job satisfaction and performance reviews, significantly improved the accuracy of predictions. They concluded that machine learning models could accurately predict employee turnover, especially when multiple relevant features are considered.

Ali proposed a hybrid model combining Random Forest and Principal Component Analysis (PCA) for employee turnover prediction (Wild Ali, 2021). PCA was used to reduce the dimensionality of the data and select the most relevant features, while Random Forest was employed to classify the employees at risk of leaving the organization. The results showed that this approach outperformed traditional models, particularly in terms of classification accuracy, and suggested that feature reduction combined with robust classification methods could yield superior performance.

Kumar et al. introduced a systematic approach to predicting employee turnover using machine learning algorithms (Kumar et al., 2023). They focused on the financial and operational implications of turnover, emphasizing the importance of retaining skilled employees to conserve organizational resources. The study tested various machine learning algorithms, including random forests, neural networks, and XGBoost, on a dataset from a multinational organization. The results showed that XGBoost achieved the highest prediction accuracy of 78.5 %, demonstrating its effectiveness in identifying high-risk employees. The study concluded that machine learning techniques can be an essential tool for HR departments to proactively address turnover and enhance workforce stability.

Shafie et al. focused on artificial neural networks (ANN) and clustering techniques to predict employee turnover (Shafie et al., 2024). They used a clustering approach to segment employees based on various features, and then applied ANN to predict the likelihood of turnover within each cluster. The findings indicated that segmenting employees before applying machine learning models improved prediction accuracy, especially when dealing with large datasets containing complex, high-dimensional data.

Biswas et al. employed ensemble learning models, specifically

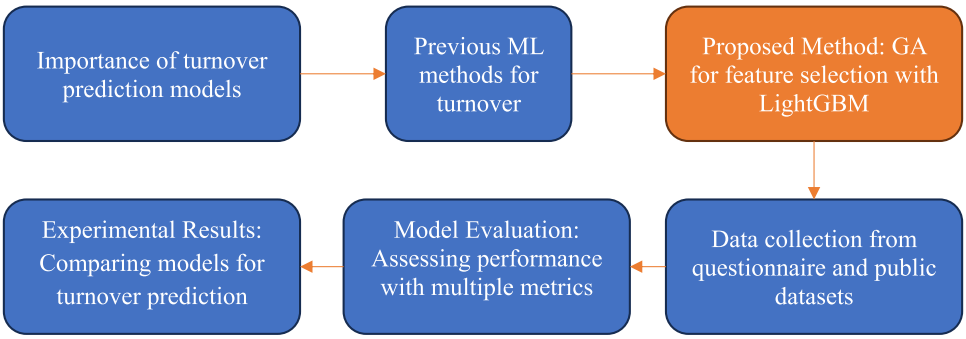


Fig. 1. Research roadmap.

combining Gradient Boosting, Random Forest, SVM, and Logistic Regression, to predict employee turnover (Biswas et al., 2023). Their results showed that ensemble models, which combine the outputs of multiple algorithms, were more accurate than individual models. This study highlighted the importance of using multiple models to account for the various factors affecting turnover, as no single model could capture all the nuances in the data.

Shankar et al. explored the challenges organizations face with high employee turnover and applied machine learning algorithms to predict employee attrition (Shankar et al., 2024). They focused on overcoming data noise in HRIS databases, which can lead to inaccurate predictions. The study utilized six machine learning algorithms to classify turnover risks, with an emphasis on decision trees, random forests, and XGBoost. The results showed that these algorithms could accurately predict voluntary turnover by analyzing historical employee data from a multinational company. Their study concluded that machine learning algorithms are effective in predicting turnover, but the issue of data noise must be addressed to improve prediction accuracy.

Chowdhury et al. focused on enhancing the transparency of AI-based machine learning models used for predicting employee turnover (Chowdhury et al., 2023). They highlighted the importance of understanding how machine learning models generate predictions to build trust among HR managers. By employing the LIME (Local Interpretable Model-Agnostic Explanations) method, they showed how HR managers could interpret the predictions generated by these models. Their findings revealed that transparency in AI models is crucial for better decision-making in HR and ensuring fairness in turnover predictions. The study suggested that transparent machine learning models can improve employee retention strategies by providing HR professionals with actionable insights.

Nayem and Uddin addressed the issue of unbiased employee performance evaluation using machine learning techniques (Nayem and Uddin, 2024). They considered various physical, social, and economic factors affecting performance, including job satisfaction, work environment, and compensation. Their findings showed that machine learning algorithms could predict employee performance and turnover by integrating multiple external factors, and that using these factors resulted in more accurate and fair performance assessments.

Veglio et al. utilized supervised machine learning techniques, particularly classification decision trees, to predict voluntary employee turnover in a multinational corporation (MNC) (Veglio et al., 2025). They applied the CHAID (Chi-square Automatic Interaction Detector) algorithm to analyze a dataset of 2932 employees. Their results showed that the decision tree model successfully identified key factors that determine whether employees would voluntarily leave the organization. Key predictors included job satisfaction, work-life balance, and organizational commitment. The study emphasized that machine learning, specifically decision trees, can provide a strategic advantage for HR departments in identifying at-risk employees and implementing targeted retention strategies.

Adeusi et al. focused on machine learning models for predicting employee turnover in high-stress sectors (Adeusi et al., 2024). Their study used a variety of machine learning algorithms, including logistic regression, decision trees, random forests, and neural networks, to predict turnover in sectors such as healthcare and finance. Their results showed that job satisfaction, stress levels, and performance ratings were the most significant predictors of turnover. They concluded that integrating machine learning models into human resource practices can help reduce turnover rates, particularly in high-stress environments.

Park et al. developed a machine learning-based model to predict turnover intention among new employees, particularly college graduates (Park et al., 2024). They used a combination of logistic regression, k-nearest neighbor (KNN), and XGBoost classifiers to analyze the factors influencing turnover, such as job security, work environment, and job preference. Their model achieved an accuracy of 78.5 %, with job security emerging as the most significant predictor of turnover. This study highlighted the importance of considering job security and organizational satisfaction in predicting turnover, particularly for new employees.

Kumar et al. proposed a systematic machine learning approach to predict employee turnover in high-stress sectors (Kumar et al., 2023). They compared various machine learning models, including random forests, neural networks, and XGBoost, to predict turnover based on job satisfaction, stress levels, and performance metrics. Their results indicated that XGBoost outperformed other models, achieving a prediction accuracy of 78.5 %. They also emphasized the need for integrating predictive models into HR practices to enhance employee retention strategies and ensure workforce stability.

In summary, although many studies have applied various machine learning techniques for employee turnover prediction, there are still critical gaps to be addressed. First, limited attention has been given to integrating powerful ensemble methods such as LightGBM with metaheuristic-based feature selection techniques like Genetic Algorithms. Second, most existing works emphasize either accuracy or model simplicity but rarely achieve both simultaneously. Finally, very few models are evaluated using both public and real-world HR datasets, reducing their generalizability. This research aims to fill these gaps by proposing a hybrid GA-LightGBM framework that balances prediction performance, model interpretability, and real-world applicability.

Table 1 provides a summary of the previous works in the field of employee turnover prediction, and Figure 2 illustrates the main challenges in the field of employee turnover prediction.

3. Proposed method

In this study, we propose a novel machine learning-based framework for predicting employee turnover by combining LightGBM with Genetic Algorithms for feature selection. This hybrid approach leverages the strengths of both techniques: LightGBM for its high predictive accuracy and Genetic Algorithms for optimizing the selection of the most relevant features for turnover prediction. Although various novel models such as deep learning or ensemble-based hybrid techniques have been proposed in recent works, we selected LightGBM due to its outstanding performance on tabular data and its computational efficiency. Furthermore, Genetic Algorithms are employed to optimize feature selection, offering enhanced model interpretability — an essential aspect in HR decision-making environments. This hybrid approach aims to balance performance and transparency, providing a practical solution that can be more readily adopted by organizations. The methodology involves the following key steps: Data Collection and Preprocessing, Feature Selection using Genetic Algorithms, Model Training using LightGBM, Model Evaluation, and Interpretation of Results.

3.1. Data collection and preprocessing

For this study, two datasets were used, both of which play a crucial role in understanding and predicting employee turnover, a critical issue in Human Resource Management. The first dataset is the HR Analytics Dataset, which is publicly available on Kaggle.[1] This dataset includes a variety of employee attributes such as job satisfaction, employee performance evaluations, salary, department, and work-life balance, making it highly relevant for analyzing factors influencing turnover. Table 2 presents the statistical characteristics of the HR dataset. The second dataset, which was internally collected, consists of data from 100 employees across different companies. It provides similar features including job role, compensation, employee satisfaction, and turnover status. The inclusion of this custom dataset adds diversity to the analysis, as it reflects real-world organizational data that can help train machine learning models for turnover prediction. Table 3 presents the statistical

[1] https://www.kaggle.com/datasets/anshika2301/hr-analytics-dataset

Table 1
Summary of Previous Works on Employee Turnover Prediction.

References	Year	Methodology	Findings	Limitations	Conclusions
Valle, M. A., Ruz, G. A., & Masías, V. H. (Valle et al. 2017)	2017	Self-organizing maps for modeling turnover in sales agents.	Identified key factors influencing turnover in sales teams, enhancing retention strategies.	Study limited to sales agents in call centers.	Self-organizing maps can provide valuable insights into employee retention in specific sectors.
Frierson, J. & Si, D. (Frierson and Si, 2018)	2018	Machine learning algorithms for evaluating attrition.	Found that machine learning models effectively predicted employee attrition using performance metrics such as ROC, recall, and precision.	Limited to specific datasets in particular industries.	Machine learning models are highly useful for attrition prediction in various sectors.
Zhao, Y., Hryniewicki, M. K., Cheng, F., Fu, B., & Zhu, X. (Zhao et al., 2019)	2019	Machine learning for employee turnover prediction.	Found that machine learning methods, such as classification algorithms, are reliable for predicting employee turnover.	Requires large datasets and continuous data refinement.	Machine learning is a reliable approach for predicting employee turnover, especially with large datasets.
de Oliveira et al. (de Oliveira et al., 2019)	2019	Gated recurrent neural networks for turnover prediction from email data.	Successfully predicted turnover using email communication patterns.	Limited generalizability due to diverse communication practices across organizations.	GRU RNNs effectively predict turnover behavior based on email data patterns.
Kodden, B., & Roelofs, J. (Kodden and Roelofs, 2019)	2019	Psychological contract as a mediator of turnover intention.	Investigated how the psychological contract influences employee turnover intention.	Limited to legal and organizational contexts.	Psychological contract is an important factor in understanding turnover intentions.
Teng et al. (Teng et al. 2019)	2019	CEHNN model integrating employee profiles and co-worker influences.	Demonstrated that turnover can be predicted by considering co-worker turnover influences (contagious effect).	Study limited to a single Chinese company.	Including contagious effects improves turnover prediction accuracy and helps retention strategies.
El-rayes, N., Smith, M., & Taylor, S. M. (El-Rayes et al. 2019)	2019	Tree-based models for employee attrition prediction.	Proposed tree-based models to predict employee attrition, showing strong predictive capabilities.	Potential bias in data used by anonymous sources.	Tree-based models are effective for turnover prediction and can be validated using ROC curves.
PM & Balaji (PM and Balaji, 2019)	2019	Weka software with various ML algorithms on a fictional IBM dataset.	Decision trees, especially J48, performed well in attrition analysis.	Naive Bayes oversimplified assumptions about feature independence.	Machine learning algorithms, including decision trees, are effective for attrition prediction, emphasizing further research.
Judrups et al. (Judrups et al. 2021)	2021	Machine learning model for predicting turnover across 67 factors.	Identified key predictors like education and business travel frequency, with 89 % precision in turnover prediction.	Limited accuracy with K-means clustering, necessitating reliance on the CART algorithm.	Machine learning models significantly aid managerial decision-making for turnover prediction.
Masoud et al. (Masoud et al. 2021)	2021	Soft-clustering unsupervised learning on employee questionnaire responses.	Identified career opportunities and working abroad as key turnover influencers.	Limited by the single organizational data source.	HR departments can reduce turnover by improving work environments and offering better incentives.
Mohamed Ahmed (Ahmed, 2021)	2021	Neural networks for turnover prediction with various classification algorithms.	Neural network showed 84 % accuracy and 74 % AUC in turnover classification.	Data limitations; additional social factors may improve model efficiency.	Neural network models effectively aid HR managers in turnover prediction, especially with factors like overtime and salary.
Kovvuri & Dommeti (Kovvuri and Dommeti, 2022)	2022	Kaggle dataset analysis with machine learning algorithms.	XGBoost outperformed other classifiers with 85.3 % accuracy.	Difficulty in identifying the most significant turnover factors.	XGBoost is the most effective algorithm for predicting employee turnover.
Atef et al. (Atef et al., 2022)	2022	KNN and RF algorithms on an IBM dataset for predicting employee turnover.	Identified higher turnover rates among employees aged 18–24.	Limited number of features; may affect model performance.	KNN outperforms RF in accuracy, precision, F-score, and specificity for turnover prediction.
Huang et al. (Huang et al. 2023)	2023	Conceptualizing personalized HRM using advanced HR analytics and AI.	Optimizing performance by analyzing key HR data in personalized HRM.	Requires advanced technology and additional administrative work.	Personalized HRM enhances HR functions and adds value beyond traditional practices.
Marquez et al. (Marquez et al., 2024)	2024	Linear regression and AI models (TensorFlow and Keras) on survey data.	Identified employee turnover factors influenced by personal and economic variables.	Complex data patterns influencing turnover.	Enhancing prediction efficiency with AI and regression for comprehensive HR management.
Khan (Rashed Khan, 2024)	2024	Systematic literature review on AI's impact in Talent Management.	Enhancing recruitment, decision-making, and employee development through AI applications.	Potential publication bias and challenges with rapid technological evolution.	Advocates for ethical and careful AI implementation in talent management.
Hasan et al. (2024))	2024	Integrated approach of business analytics and machine learning for employee performance prediction.	Demonstrated the effectiveness of this method for improving employee performance prediction.	Necessity for periodic model updates to adapt to changing business environments.	This method improves organizational decision-making and performance management.
Taqatqa, et al. (Taqatqa, Taqatqa, Sara, Samhan, and Kanan, 2025)	2025	CRISP methodology with machine learning classification models (Random Forest, Logistic Regression, SVM)	Random Forest emerges as the most effective model for employee turnover prediction.	Study focused on a specific university dataset, limiting generalization	Machine learning, especially Random Forest, can enhance the effectiveness of turnover predictions.
Bustillo, J. C. M. (Bustillo, 2025)	2025	Optimization-based prediction model with Genetic Algorithms and KNN/Naive Bayes.	The FMSAX operator paired with KNN achieves the highest predictive accuracy, precision, F-measure, and recall.	Focuses only on specific operators and features, not a broad range of machine learning techniques.	GA-optimized models significantly improve turnover prediction and offer actionable retention insights.

(*continued on next page*)

Table 1 (*continued*)

References	Year	Methodology	Findings	Limitations	Conclusions
Veglio, et at. (Veglio et al., 2025)	2025	Supervised machine learning with classification decision trees.	Identified determinants of voluntary employee turnover in a multinational telco corporation, highlighting the potential of classification trees.	The study focuses on a single multinational corporation, which may limit generalizability to other industries.	Classification decision trees offer a strategic approach to identifying employee turnover risks, enabling organizations to develop targeted retention strategies.

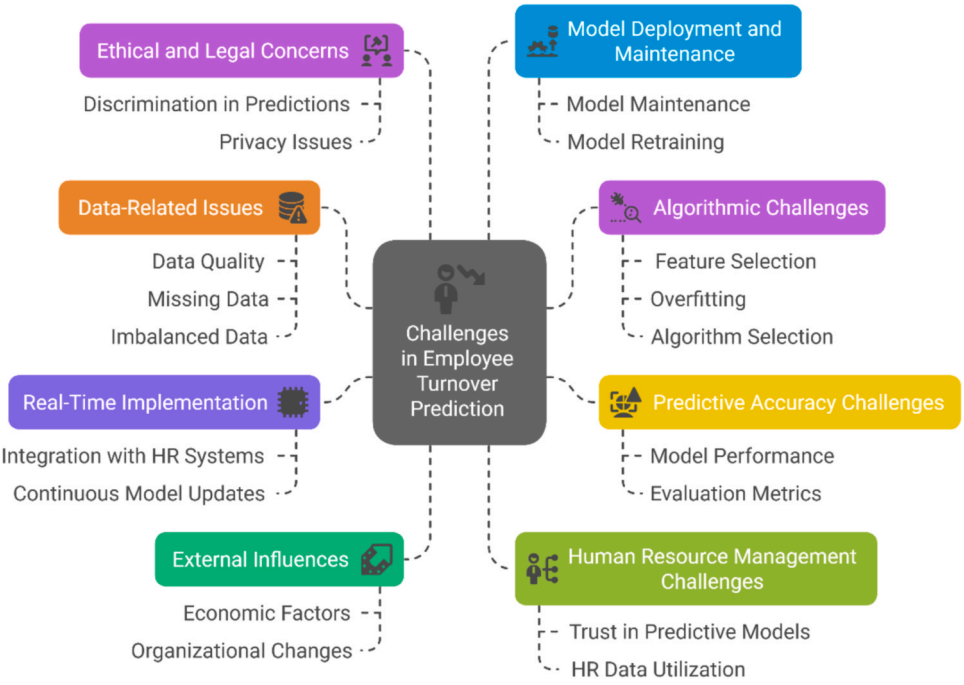


Fig. 2. Key Challenges in Employee Turnover Prediction.

Table 2
Statistical Characteristics of the HR Analytics Dataset.

Variable	Min	Max	Mean	Std Deviation
Age	18	60	36.92	9.13
DailyRate	102	1499	801.38	403.13
DistanceFromHome	1	29	9.22	8.13
Education	1	5	2.91	1.02
EmployeeCount	1	1	1	0
EmployeeNumber	1	2068	1031.86	605.96
EnvironmentSatisfaction	1	4	2.72	1.09
HourlyRate	30	100	65.85	20.33
JobInvolvement	1	4	2.73	0.71
JobLevel	1	5	2.06	1.11
JobSatisfaction	1	4	2.72	1.1
MonthlyIncome	1009	19999	6504.99	4700.26
MonthlyRate	2094	26999	14298.46	7112.06
NumCompaniesWorked	0	9	2.69	2.49
PercentSalaryHike	11	25	15.21	3.66
PerformanceRating	3	4	3.15	0.36
RelationshipSatisfaction	1	4	2.71	1.08
StandardHours	80	80	80	0
StockOptionLevel	0	3	0.79	0.85
TotalWorkingYears	0	40	11.28	7.77
TrainingTimesLastYear	0	6	2.8	1.29
WorkLifeBalance	1	4	2.76	0.71
YearsAtCompany	0	40	7.01	6.12
YearsInCurrentRole	0	18	4.23	3.62
YearsSinceLastPromotion	0	15	2.18	3.22
YearsWithCurrManager	0	17	4.12	3.56

Table 3
Statistical Characteristics of the Organizaiton Dataset.

Variable	Min	Max	Mean	Standard Deviation
Age	22	58	39.53	10.13
YearsAtCompany	0	40	18.22	11.52
Salary	25452	119796	78271.99	28861.34
JobSatisfaction	1	4	2.38	1.11
WorkLifeBalance	1	4	2.47	1.16
EnvironmentSatisfaction	1	4	2.5	1.1
JobInvolvement	1	4	2.65	1.09
EducationLevel	1	5	3.1	1.27
PerformanceRating	1	4	2.47	1.18
NumberOfCompaniesWorked	0	9	4.54	2.82
YearsInCurrentRole	0	15	7.94	4.67
YearsSinceLastPromotion	0	10	5.39	2.97
StockOptionLevel	0	3	1.39	1.1
TotalWorkingYears	1	40	19.57	11.86
DistanceFromHome	1	39	20.63	12.07
EmployeeID	1	5000	2500	1444
Gender	Male	Female	Female	-
JobRole	Sales	Engineer	Engineer	-
OverTime	No	Yes	Yes	-

characteristics of the Organization dataset. The datasets contain both categorical (e.g., job role, department) and continuous variables (e.g., salary, satisfaction level), and are preprocessed as follows:

Handling Missing Data: Missing values are addressed by using imputation techniques such as filling with the mean or mode of the respective column.

Feature Encoding: Categorical variables (e.g., department, education

level) are encoded using One-Hot Encoding.

Feature Scaling: Continuous variables like salary and satisfaction are scaled using Min-Max scaling to normalize the values and ensure uniformity in input ranges.

Outlier Detection: Z-score and IQR-based methods are used to detect and handle outliers that may distort the model's performance.

To assess multicollinearity among independent variables, we computed Pearson correlation coefficients and visualized the relationships using heatmaps for both datasets (Figure 7). Features with high correlation (>0.8) were flagged and considered for removal or combination. Additionally, we ensured that the same preprocessing pipeline was applied consistently to both datasets, covering missing data handling, encoding, scaling, and outlier detection.

3.2. Genetic algorithm

Feature selection is a critical step for improving the efficiency of machine learning models (Xia et al., 2015). To optimize the selection of the most relevant features for employee turnover prediction, Genetic Algorithms are employed. The Genetic Algorithm is a popular optimization technique inspired by the process of natural selection. It is widely used for feature selection in machine learning models due to its ability to effectively search large, high-dimensional feature spaces. In the initialization phase, the GA generates an initial population of potential solutions, which are feature subsets. These subsets are created randomly, where each individual in the population represents a set of features selected from the original dataset.

$$\text{Population} = [X_1, X_2, \quad \ldots, X_N] \tag{1}$$

Where each X_i represents a feature subset, and N is the total population size. Each individual (feature subset) in the population is evaluated for its fitness based on the performance of a machine learning model (e.g., LightGBM) trained using that subset. The fitness of an individual is typically measured by the prediction accuracy or other relevant performance metrics (such as F1-score, precision, recall, etc.).

$$\text{Fitness}(X_i) = \text{Accuracy}(X_i) \tag{2}$$

Where $\text{Accuracy}(X_i)$ is the performance of the machine learning model trained on feature subset X_i. In the selection step, feature subsets with better performance (higher fitness) are selected to create the next generation. The probability of selection for each individual is proportional to its fitness.

$$P(X_i) = \frac{\text{Fitness}(X_i)}{\sum_{i=1}^{N} \text{Fitness}(X_i)} \tag{3}$$

Where $P(X_i)$ is the probability of selecting feature subset X_i, and the sum of all fitness values normalizes the probabilities. During crossover, two parent feature subsets are combined to produce offspring. This mimics the biological crossover process, where parts of two parent genomes are exchanged to create a new individual. In the case of feature selection, this step involves combining feature subsets from two parent individuals to form a new subset. The crossover mechanism ensures diversity in the population by mixing features from different solutions. Mutation introduces small random changes to an offspring's feature subset to explore new parts of the solution space. This is akin to genetic mutations in biological organisms and helps maintain diversity within the population. Mutation can either add a new feature to the subset or remove an existing feature. The GA algorithm continues to evolve the population through selection, crossover, and mutation for several generations. The process terminates when a specified number of generations have been completed or when the fitness of the population reaches a predefined threshold (indicating convergence).

In our implementation, the initial population of feature subsets (Equation (1)) was generated using all available features (30 for HR dataset, 25 for Organization). The fitness function in Equation (2) was evaluated using LightGBM performance on selected feature subsets, and Equation (3) governed selection probability based on classification accuracy using 10-fold cross-validation.

By iteratively evolving the population of feature subsets, the genetic algorithm seeks to retain the most important features for predicting employee turnover, while discarding less informative or redundant features. This process ensures that the resulting feature set is optimized for the machine learning model, leading to improved prediction accuracy and reduced overfitting. Figure 3 illustrates the structure of the genetic algorithm, highlighting the key stages of initialization, selection, crossover, mutation, and termination.

In this study, the Genetic Algorithm was applied separately to the HR Analytics and Organization datasets. Initially, the HR dataset included 30 features, from which the GA selected 14 optimal features. For the Organization dataset, 25 initial features were reduced to 11 after GA-based selection. These include demographic (e.g., age, education), job-related (e.g., role, department, salary), and performance-based (e.g., evaluation scores, overtime, satisfaction levels) variables. A full list of features used for each dataset is presented in Table 4. The GA was configured with a population size of 50, a maximum of 100 generations, a crossover rate of 0.8, and a mutation rate of 0.1. The fitness function was based on 10-fold cross-validated F1-score using LightGBM. This feature reduction led to more interpretable models and reduced overfitting risk.

In the Roulette Wheel selection process, each feature subset is assigned a selection probability proportional to its fitness value. The fitness of each subset is evaluated based on the performance of a LightGBM model trained with the subset of features. The selection probability for each individual is calculated as:

$$P(i) = \frac{fitness(i)}{\sum fitness(i)} \tag{4}$$

where $P(i)$ is the probability of selecting individual i. A random number is then generated, and the individual whose probability corresponds to that number is selected for crossover in the next generation.

3.2.1. Genetic algorithm parameters and operators

To enhance the predictive power and reduce feature redundancy, a Genetic Algorithm was employed for feature selection. The algorithm was configured with the following settings and operators:

The initial population consists of 50 individuals, each represented by a binary chromosome. The length of each chromosome corresponds to the total number of input features (30 for the HR dataset, 25 for the Organization dataset). Each gene is initialized randomly, where a value

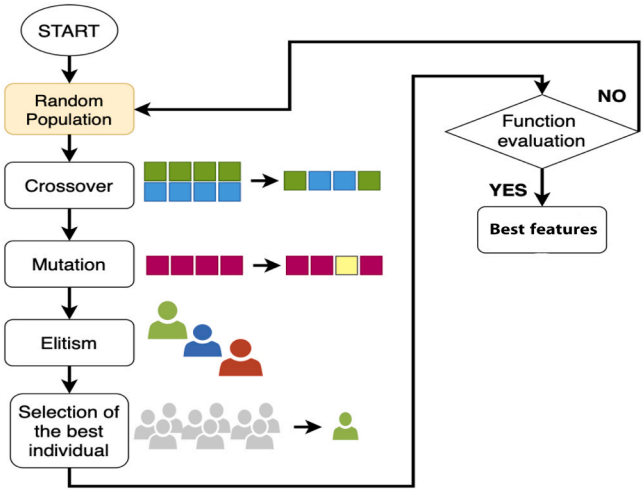


Fig. 3. Structure of the Genetic Algorithm (Mirjalili and Mirjalili, 2019).

Table 4
Input Features Considered for Genetic Algorithm from Both Datasets.

Feature Name	Description	Dataset Source	Type
Age	Employee's age	Both	Continuous
MonthlyIncome / Salary	Monthly salary (HR) or annual salary (Org)	Both	Continuous
JobSatisfaction	Job satisfaction rating	Both	Categorical
WorkLifeBalance	Work-life balance score	Both	Categorical
EnvironmentSatisfaction	Satisfaction with work environment	Both	Categorical
JobInvolvement	Employee's job involvement level	Both	Categorical
EducationLevel / Education	Education level	Both	Categorical
PerformanceRating	Performance evaluation	Both	Categorical
YearsAtCompany	Years of service in the company	Both	Continuous
YearsInCurrentRole	Time spent in current role	Both	Continuous
YearsSinceLastPromotion	Time since last promotion	Both	Continuous
TotalWorkingYears	Total work experience	Both	Continuous
DistanceFromHome	Distance between home and workplace	Both	Continuous
StockOptionLevel	Stock options level	Both	Categorical
NumCompaniesWorked	Number of companies previously worked at	Both	Continuous
Gender	Gender of the employee	Organization Dataset	Categorical
OverTime	Whether the employee works overtime	Organization Dataset	Categorical
JobRole	Role/position in the organization	Organization Dataset	Categorical
Department	Employee's department	HR Analytics Dataset	Categorical
DailyRate	Daily salary rate	HR Analytics Dataset	Continuous
HourlyRate	Hourly pay	HR Analytics Dataset	Continuous
MonthlyRate	Monthly rate before tax	HR Analytics Dataset	Continuous
TrainingTimesLastYear	Trainings attended last year	HR Analytics Dataset	Continuous
StandardHours	Standard working hours	HR Analytics Dataset	Constant (80)
EmployeeNumber / ID	Unique ID for employee	Both	Identifier
EmployeeCount	Constant count (not informative)	HR Analytics Dataset	Constant (1)

of '1' denotes that the feature is selected, and '0' denotes exclusion.

The fitness of each individual is evaluated using the F1-score of a LightGBM classifier trained on the selected features. A 10-fold cross-validation scheme is used to ensure reliable evaluation and avoid overfitting.

Roulette Wheel Selection is used to choose parent chromosomes for reproduction. The probability of selection is proportional to each individual's fitness score, thus favoring better-performing solutions while maintaining diversity.

Uniform crossover is applied with a crossover rate of 0.8. In this operator, genes from two parent chromosomes are combined such that each gene in the offspring is independently selected from one of the two parents with equal probability.

A bit-flip mutation is used to introduce variation into the population. Each gene in the chromosome has a 10 % probability of being flipped from 0 to 1 or vice versa. This helps the algorithm avoid local optima and explore new feature subsets.

The top 5 individuals (10 % of the population) with the highest fitness scores are directly carried over to the next generation. This ensures that the best solutions are preserved and not lost due to crossover or mutation.

The algorithm runs for a maximum of 100 generations or stops early if no improvement in the best fitness score is observed for 10 consecutive generations.

These settings allow the GA to effectively search the high-dimensional feature space and retain only the most informative attributes for turnover prediction.

3.3. LightGBM

After feature selection, LightGBM is used to train the turnover prediction model. LightGBM is an advanced, efficient, and scalable implementation of gradient boosting, designed to handle large datasets and high-dimensional data efficiently (X. Sun et al., 2020). It leverages histogram-based learning, where continuous feature values are discretized into bins, reducing computational costs by simplifying the search for optimal split points. Unlike traditional gradient boosting methods, which grow trees level-wise, LightGBM uses a leaf-wise growth strategy, selecting the leaf that maximizes the reduction in the loss function at each step (Ke et al., 2017). This often results in deeper trees and better predictive accuracy. The framework can natively handle categorical features without the need for explicit encoding, making it more efficient than other models that require pre-processing like one-hot encoding. LightGBM is optimized for speed, memory efficiency, and scalability, making it capable of handling massive datasets with millions of samples and features. It also supports parallel and distributed training, which accelerates the model-building process on large-scale data. Key hyperparameters in LightGBM include `num_leaves` (the maximum number of leaves in a tree), `max_depth` (the depth limit of the tree to avoid overfitting), and `learning_rate` (to control the speed of learning). The framework also includes features such as `subsample` for row sampling and `colsample_bytree` for feature sampling, both of which help prevent overfitting (D. Zhang and Gong, 2020). LightGBM's ability to handle large datasets and its focus on speed and memory efficiency make it one of the most popular choices for machine learning tasks like classification, regression, and ranking. It has been widely adopted across industries such as finance, healthcare, e-commerce, and natural language processing (NLP), where it has been used for tasks ranging from fraud detection and credit scoring to customer churn prediction and text classification. The primary goal in LightGBM is to optimize an objective function, which typically consists of two main components: the loss function and the regularization term. For a decision tree model, the objective function can be written as (Liao et al., 2022):

$$L(\theta) = \sum_{i=1}^{n} l(y_i, f(x_i)) + \Omega(f) \tag{5}$$

Where $l(y_i, f(x_i))$ is the loss function that measures the difference between the true value y_i and the prediction $f(x_i)$.

$\Omega(f)$ is the regularization term that penalizes the complexity of the model (to avoid overfitting). At each stage of training, LightGBM uses gradient descent to update the model. The model is iteratively optimized by minimizing the loss function using gradient updates. Specifically, the new model at each step is:

$$f_{t+1} = f_t + \eta . h_t \tag{6}$$

Where:

- f_t is the current model.
- h_t is the new tree that reduces the error of the model.

- η is the learning rate, controlling the speed of updates.

LightGBM grows trees in a leaf-wise manner, which means instead of growing the tree level-wise, it grows the tree leaf-wise by choosing the leaf with the maximum reduction in the loss function at each step. This strategy generally results in deeper trees and thus better predictive performance. The tree grows by finding the best split using criteria such as Information Gain or Gini Impurity. LightGBM can directly handle categorical features. These features are represented as integers, and the algorithm automatically finds the best splits for these features. This eliminates the need for explicit preprocessing like one-hot encoding, reducing data transformation time. For updating the model, LightGBM directly uses the gradient of the loss function. Specifically, for each step ttt, the gradient for the samples is calculated as (Wang et al., 2022):

$$g_i^{(t)} = \frac{\partial L(y_i, f(x_i))}{\partial f(x_i)} \tag{7}$$

Where:

- $g_i^{(t)}$ is the gradient of the loss function for sample i at step t.
- $L(y_i, f(x_i))$ is the loss function.

These gradients are used to construct the new decision tree, which minimizes the model's error. Figure 4 shows the schematic diagram of the LightGBM method.

These equations were applied during LightGBM model training using the selected features from GA. For the HR dataset, 14 features were used; for the Organization dataset, 11 features were selected. Loss gradients (Equation (7)) and split gains were computed accordingly.

Key Parameters of LightGBM:

num_leaves: The maximum number of leaves in each tree. This parameter directly influences model accuracy and the complexity of the trees.

max_depth: The maximum depth of the trees, used to avoid overfitting by limiting the tree's growth.

learning_rate: Controls the speed of learning, determining how quickly the model adapts during training.

subsample and colsample_bytree: These parameters randomly sample data points and features for each tree, helping to prevent overfitting.

LightGBM uses a histogram-based learning approach, which discretizes continuous feature values into bins, reducing computational cost by simplifying the search for optimal split points. Unlike traditional gradient boosting methods that grow trees level-wise, LightGBM employs a leaf-wise growth strategy, where the algorithm selects the leaf that maximizes the reduction in the loss function at each step. This approach often results in deeper trees and better predictive performance. After feature selection by the GA, the selected features are used to train the turnover prediction model using LightGBM. The model is fine-tuned using hyperparameters optimized through grid search, ensuring the best possible performance on the data.

The LightGBM classifier was trained separately on the feature subsets selected by GA from each dataset. We used the selected 14 features from the HR dataset and 11 features from the Organization dataset to train and validate the LightGBM model using 10-fold cross-validation. The key hyperparameters (e.g., num_leaves = 31, max_depth = 10, learning_rate = 0.1) were optimized using grid search. The results showed that reducing input features not only improved model generalization but also significantly decreased training time.

3.4. Machine learning-based framework

The integration of Genetic Algorithms and LightGBM for feature selection combines the global search capabilities of GA with the powerful predictive performance of LightGBM. This hybrid approach aims to select the optimal feature subset that maximizes model performance while minimizing computational complexity. The process starts by encoding feature subsets as chromosomes. In the context of genetic algorithms for feature selection, each potential solution is represented as a chromosome. A chromosome is a binary string where each bit corresponds to a feature in the dataset. The bit value of '1' indicates that the feature is selected for the model, while '0' means the feature is excluded. This binary encoding allows the GA to explore different subsets of features efficiently. Mathematically, a chromosome C representing a feature subset for n features is defined as:

$$C = [f_i] \ (i = 1,\ 2, \ldots, n) \tag{8}$$

Where:

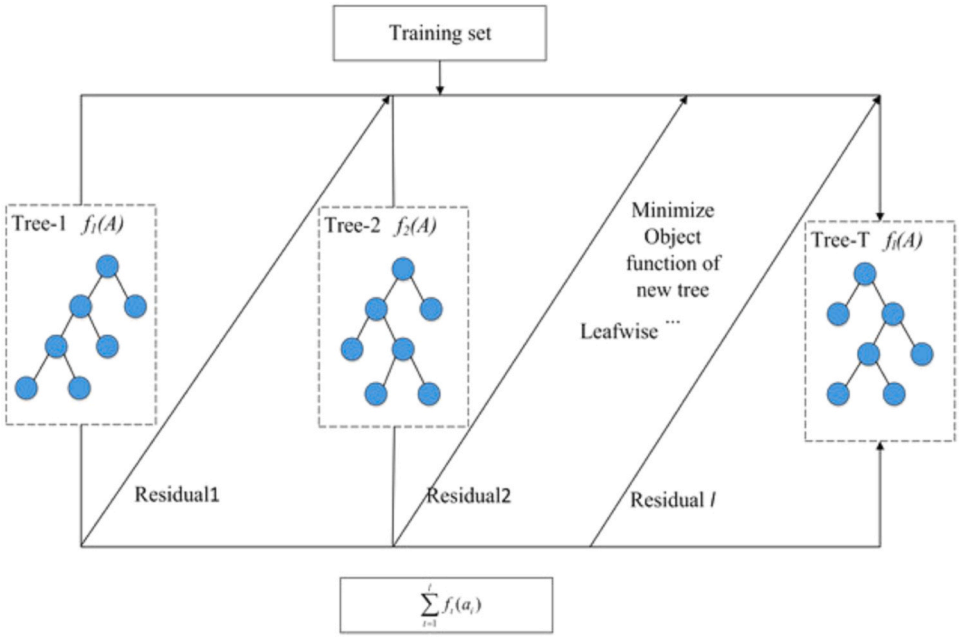


Fig. 4. Schematic Diagram of the LightGBM Method (Wang et al., 2022).

- $f_i \epsilon\ 0, 1$, representing whether feature i is included in the model ($f_i = 1$) or excluded ($f_i = 0$).

- n is the total number of features in the dataset.

This encoding allows the GA to search through the feature space by selecting and rejecting features through genetic operations. The fitness function in a genetic algorithm is used to evaluate the quality of each solution (chromosome). In the case of feature selection, the fitness of each chromosome is determined by training a LightGBM on the selected features (i.e., the features corresponding to the 1-bits in the chromosome). The performance of the LightGBM, which could be measured in terms of accuracy, precision, recall, or AUC, serves as the fitness score for the chromosome. Given a chromosome C, the fitness $F(C)$ can be expressed as:

$$F(C) = \textit{Performance Metric}(\textit{Model trained on features selected by } C) \quad (9)$$

Where the LightGBM is trained only on the features selected by the binary representation of the chromosome. The goal is to maximize the fitness function, i.e., to maximize the LightGBM's predictive performance while minimizing the number of selected features.

The process of selection, crossover, and mutation continues for several generations, evolving the population toward the optimal feature set. The chromosome with the highest fitness represents the optimal subset of features. These selected features are then used to train the final LightGBM. The objective of combining Genetic Algorithms with LightGBM for feature selection is to optimize the set of features that will maximize the model's performance while minimizing complexity. The optimization process is formulated as a maximization of the objective function, which combines the model's performance (as evaluated by LightGBM) with a regularization term to prevent overfitting. The objective function $L(C)$ can be expressed as:

$$\textit{Maximize } F(C) = \frac{1}{n} \times \sum_{i=1}^{n} l(y_i, f(x_i, C)) + \lambda \times \Omega(C) \quad (10)$$

Where:

- $l(y_i, f(x_i, C))$ is the loss function that measures the error between the true value y_i and the predicted value $f(x_i, C)$ using the selected features.
- $\Omega(C)$ is the regularization term that penalizes the model for selecting too many features, ensuring that the feature set remains minimal.
- λ is the regularization parameter that controls the trade-off between model performance and complexity.

By optimizing this objective function, the GA selects the feature subset that maximizes the predictive accuracy of the LightGBM while minimizing the complexity of the selected feature set, thus preventing overfitting and ensuring better generalization. This combination leverages the strengths of both GA and LightGBM, enabling the selection of a reduced set of highly informative features while avoiding overfitting and improving model accuracy. Figure 5 summarizes the diagram of the proposed method. Initially, a binary chromosome was generated for each individual in the population, where each gene represented the inclusion (1) or exclusion (0) of a feature from the dataset. The initial population was created using all features (30 for HR dataset, 25 for Organization). Each chromosome's fitness was evaluated by training a LightGBM model using only the selected features and computing its F1-score with 10-fold cross-validation. Based on fitness values, the top-performing chromosomes were selected using roulette wheel selection. These were then used in crossover (one-point) and mutation operations to generate the next generation. The process repeated for 100 generations or until convergence.

The proposed hybrid model was implemented using MATLAB 2020 on a system with an Intel 2.7 GHz processor and 12 GB RAM. The implementation consisted of two main components: the Genetic Algorithm for feature selection and LightGBM for classification.

– Initialization: The Genetic Algorithm begins by generating a random initial population. Each individual in the population is represented as

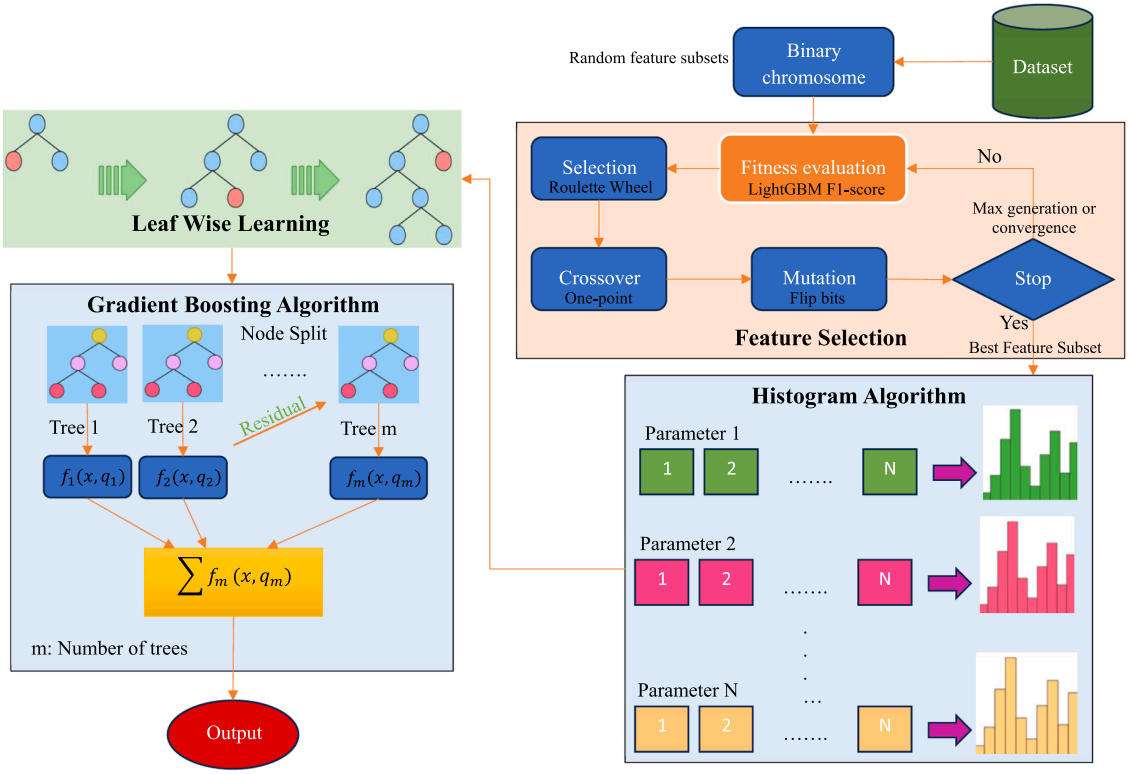


Fig. 5. The diagram of the proposed method for predicting employee turnover.

a binary chromosome of length n, where n is the number of features. A bit value of 1 indicates that the feature is selected; 0 means it is excluded.
- Fitness Evaluation: Each individual (feature subset) is evaluated by training a LightGBM classifier using only the selected features. The fitness function is defined as the model's F1-Score on 10-fold cross-validation, ensuring stability and robustness in evaluation.
- Selection: We used roulette wheel selection to choose parents based on their relative fitness. Individuals with higher F1-scores had a higher probability of being selected for crossover.
- Crossover: Crossover was performed using uniform crossover, where a new child is generated by randomly mixing bits from two parents. This maintains genetic diversity and allows exploration of new feature combinations.
- Mutation: To avoid premature convergence, mutation was applied by flipping a small percentage (typically 5 %) of the bits in selected individuals. This helps the algorithm explore new areas of the feature space.
- Replacement & Iteration: The next generation is formed by selecting the best individuals from both parents and offspring, ensuring elitism. This process repeats for a predefined number of generations (e.g., 50 iterations) or until convergence (i.e., no improvement in fitness for 10 consecutive generations).
- Final Model Training: Once the optimal feature subset is found, a LightGBM model is trained using the selected features on the full training set. The hyperparameters of LightGBM (e.g., *num_leaves*, *learning_rate*, *max_depth*) were set based on grid search and prior research (as listed in Table 12).

This implementation pipeline ensures that only the most informative features are passed to the classifier, resulting in improved accuracy, reduced overfitting, and better interpretability.

The fitness function evaluates the performance of each feature subset by training a LightGBM model using the selected features. The evaluation uses standard classification metrics such as accuracy, precision, recall, and F1-score, computed through 10-fold cross-validation."

The hyperparameters of LightGBM, including num_leaves, max_depth, and learning_rate, are evaluated using the features selected by the GA. The performance is assessed based on accuracy and AUC, ensuring that the model does not overfit the selected features but instead generalizes well across the dataset.

4. Model evaluation

In machine learning, performance metrics are crucial for evaluating the effectiveness of a model. To evaluate the performance of the model, we use several common metrics to assess its predictive power.

4.1. Performance criteria

A confusion matrix is a crucial tool in data mining for evaluating the performance of models and machine learning algorithms (Muntean and Militaru, 2023). It displays the count of correct and incorrect predictions for each class by comparing the actual values (labeled samples) with the predictions made by the model. A confusion matrix is structured as an $n * n$ table, where n represents the number of classes in the problem. The actual classes are placed along the vertical axis, and the predicted classes are placed along the horizontal axis of the table. Typically, a confusion matrix is presented as shown in Figure 6.

The values that are placed in the cells of the confusion matrix are as follows:

- True Positive (*TP*): The number of samples that are correctly identified and belong to the positive class.
- True Negative (*TN*): The number of samples that are correctly identified and belong to the negative class.

		Predicted	
		Positive	Negative
Actual	Positive	TP	FN
	Negative	FP	TN

Fig. 6. Confusion matrix (S. Zhang et al., 2003).

- False Positive (*FP*): The number of samples that are incorrectly identified and actually belong to the negative class.
- False Negative (*FN*): The number of samples that are incorrectly assigned to the negative class and actually belong to the positive class.

Using the confusion matrix, we can calculate metrics such as *Accuracy*, *Precision*, *Recall* (*Sensitivity), and F1 score* of the model as follows (Juvitayapun, 2021):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

$$Precision = \frac{TP}{TP + FP} \tag{12}$$

$$Recall = \frac{TP}{TP + FN} \tag{13}$$

$$F1 = \frac{Precision.Recall}{Precision + Recall}.2 \tag{14}$$

$$AUC = \int_0^1 TPR\ d(FPR) \tag{15}$$

These metrics provide insights into different aspects of the model's performance and help evaluate its effectiveness in classification tasks. Accuracy is not always the best metric for imbalanced datasets so we use metrics such as sensitivity, specificity to better evaluate the model performance. The F1 Score is the harmonic mean of Precision and Recall. It is useful when we need to balance both precision and recall. The ROC curve is a graphical representation of the trade-off between true positive rate (Recall) and false positive rate (FPR) for different threshold values. The area under the ROC curve (AUC) is used to measure the model's ability to distinguish between classes. The closer the AUC is to 1, the better the model is at distinguishing between the classes. Additionally, we used the 10-fold cross-validation method for model evaluation, which prevents biased results (Z. Sun et al., 2024).

4.2. Algorithms

4.2.1. Support vector machine

SVM is a supervised learning algorithm used for classification and regression. It finds the hyperplane that best separates the classes by maximizing the margin between data points of different classes. It works well for both linear and non-linear data with the help of kernel functions (Tanveer et al., 2024). Table 5 shows the SVM algorithm step by step.

Table 5
SVM Algorithm.

Step	Description
1. Initialization	– Input: (x, y), $x \in \mathbb{R}^n$, $y \in -1, +1$. – Initialize w, b.
2. Calculate Margin	- Maximize margin $1/2 * \|\|w\|\|^2$ subject to $y_i(w \bullet x_i + b) \geq 1$.
3. Optimization	- Solve min $1/2 * \|\|w\|\|^2$ subject to $y_i(w \bullet x_i + b) \geq 1$.
4. Prediction	– Predict: $f(x) = w \bullet x + b$. – Classify: $y = sign(f(x))$.

4.2.2. Logistic regression

Logistic Regression is a statistical model used for binary classification problems. It predicts the probability of a binary outcome by applying the logistic (sigmoid) function to a linear combination of input features. The output is a value between 0 and 1 (Zabor et al., 2022). Table 6 shows the LR algorithm step by step.

4.2.3. Random forest

Random Forest is an ensemble learning method that creates a collection of decision trees during training and outputs the class that is the majority vote of the individual trees. It reduces overfitting and improves the accuracy of predictions (Z. Sun et al., 2024). Table 7 shows the Random Forest algorithm step by step.

4.2.4. K-nearest neighbour

KNN is a non-parametric algorithm used for classification and regression. In this algorithm, the class of a new sample is predicted based on the k nearest neighbors from the training data. Distances are typically calculated using metrics like Euclidean distance (Cerulli, 2023). Table 8 shows the KNN algorithm step by step.

4.2.5. Decision tree

Decision Tree is a supervised learning algorithm used for classification and regression. It splits the data into subsets based on feature values, creating a tree-like structure where each node represents a decision rule. The tree is constructed by recursively partitioning the data (Costa and Pedreira, 2023). Table 9 shows the Decision algorithm step by step.

4.2.6. Naive bayes

Naive Bayes is a probabilistic classifier based on Bayes' Theorem. It assumes that the features are conditionally independent given the class. Despite the simplicity, it performs well on large datasets, especially in text classification tasks (Wickramasinghe and Kalutarage, 2021). Table 10 shows the Naive Bayes algorithm step by step.

4.2.7. Extreme gradient boosting

XGBoost is an optimized version of the gradient boosting algorithm. It uses decision trees as base learners and applies boosting techniques to improve performance by combining weak learners sequentially. It is efficient, scalable, and often used for structured/tabular data (Jain and Nayyar, 2018). Table 11 shows the XGBoost algorithm step by step.

4.3. Experimental design

In this section, the initial settings for the machine learning algorithms used in this study are outlined. These settings were carefully selected through a process of trial and error, taking into account previous research and best practices from similar studies (Al Akasheh et al., 2024; Chakraborty et al., 2021; Juvitayapun, 2021; Kumar et al., 2023). By experimenting with different configurations, we aimed to identify the optimal parameters for each algorithm to ensure accurate and reliable predictions. Table 12 displays the selected settings for seven different machine learning methods, including KNN, SVM, Random Forest, XGBoost, Decision Tree, Logistic Regression, and Naive Bayes. The

Table 6
Logistic Regression Algorithm.

Step	Description
1. Initialization	– Input: (x, y), $x \in \mathbb{R}^n$, $y \in 0, 1$. – Initialize weights w and bias b.
2. Sigmoid Function	- Compute sigmoid function: $\sigma(z) = 1 / (1 + \exp(-z))$, where $z = w \bullet x + b$.
3. Compute Cost Function	- Compute the cost: $J(w, b) = -(1/m) \Sigma [y_i \log(\sigma(z_i)) + (1 - y_i) \log(1 - \sigma(z_i))]$.
4. Gradient Descent	- Update weights and bias: $w = w - \alpha * \nabla J(w)$, $b = b - \alpha * \nabla J(b)$, where α is the learning rate.
5. Prediction	– Predict: $y = \sigma(w \bullet x + b)$. – Classify: $y = 1$ if $\sigma(w \bullet x + b) > 0.5$, else $y = 0$.

Table 7
Random Forest Algorithm.

Step	Description
1. Initialization	– Input: (x, y), $x \in \mathbb{R}^n$, $y \in 0, 1$ or continuous values. – Define the number of trees (T) and the number of features (m) to use for each split.
2. Bootstrap Sampling	- For each tree, create a bootstrap sample by randomly sampling with replacement from the training data.
3. Build Decision Trees	- For each bootstrap sample, build a decision tree using random subsets of features at each split. Split using the best feature to minimize impurity (e.g., Gini or Entropy).
4. Tree Growth	- Continue splitting the data until a stopping criterion is met (e.g., max depth, minimum samples per leaf).
5. Prediction	– For classification: Each tree votes on the class, and the majority vote is taken as the final prediction. – For regression: The average prediction of all trees is used.

Table 8
K-Nearest Neighbour Algorithm.

Step	Description
1. Initialization	– Input: (x, y), $x \in \mathbb{R}^n$, $y \in 0, 1$ or continuous values. – Define number of nearest neighbors k (typically an odd number to avoid ties).
2. Compute Distance	- For each test sample, compute the distance between the test sample x and all training samples x_i using a distance metric (e.g., Euclidean distance).
3. Find Nearest Neighbors	- Select the k nearest neighbors based on the smallest distance values.
4. Majority Voting	- For classification: The class label is determined by the majority vote of the k nearest neighbors.
5. Prediction	- For regression: The output is the average value of the k nearest neighbors. For classification: The output is the majority class among the k neighbors.

Table 9
Decision Tree Algorithm.

Step	Description
1. Initialization	– Input: (x, y), $x \in \mathbb{R}^n$, $y \in 0, 1$ or continuous values. – Define parameters: max depth, min samples per split, etc.
2. Compute Best Split	- For each feature, calculate the best split based on a splitting criterion (e.g., Gini index, Information Gain).
3. Split the Data	- Partition the dataset into subsets based on the best split found in the previous step.
4. Recursive Tree Growth	- Repeat the splitting process recursively for each subset until a stopping criterion is met (e.g., max depth, min samples per leaf).
5. Prediction	- For classification: Assign the class that appears most often in the leaf node.- For regression: Take the average of the target values in the leaf node.

Table 10
Naive Bayes Algorithm.

Step	Description
1. Initialization	– Input: (x, y), $x \in \mathbb{R}^n$, $y \in 0, 1$ or categorical classes. – Define the probability distributions for each feature.
2. Compute Prior Probabilities	- Calculate the prior probability for each class: $P(y) = \text{count}(y) / \text{total_samples}$.
3. Compute Conditional Probabilities	- For each feature, compute the conditional probability $P(x_i \mid y)$, the probability of feature x_i given class y.
4. Apply Bayes' Theorem	- For a test sample, apply Bayes' theorem to compute the posterior probability for each class: $P(y\|x) \propto P(y) * \prod P(x_i \mid y)$.
5. Prediction	- For classification: Assign the class with the highest posterior probability as the predicted class.

Table 11
XGBoost Algorithm.

Step	Description
1. Initialization	– Input: (x, y), $x \in \mathbb{R}^n$, y ∈ 0, 1 or continuous values. – Initialize the base model and set the number of boosting rounds (iterations).
2. Compute Gradient	- For each iteration, compute the gradient (negative gradient) of the loss function with respect to the current model's prediction.
3. Fit a Base Learner	- Fit a new weak learner (typically a decision tree) to the residual errors from the previous iteration.
4. Update the Model	- Update the model by adding the newly fitted base learner with a learning rate to improve predictions.
5. Repeat for All Iterations	- Repeat steps 2–4 for a defined number of iterations or until convergence.- The final model is a weighted sum of all base learners.
6. Prediction	– For classification: Apply a decision threshold on the final model to make class predictions. – For regression: Use the output of the final model as the predicted value.

Table 12
Selected Settings for Machine Learning Algorithms.

Algorithm	Initial Settings
K-Nearest Neighbour	*k*: 5*Distance Metric*: Euclidean*Weighting*: Distance-based
Support Vector Machine	*Kernel*: RBF*C*: 1γ: 0.01
Logistic Regression	*C*: 1.0*Solver*: 'liblinear'*Penalty*: 'l2'
Random Forest	$n_{estimators}$:100*max_depth*: 10*min_samples_split*: 2
Decision Tree	*max_depth*: 10*min_samples_split*: 2.criterion: 'gini'.
Naive Bayes	*Alpha*: 1.0*fit_prior*: True.
XGBoost	$n_{estimators}$: 100*learning_rate*: 0.1*Normalizaion P.: 0.01max_depth*: 6subsample: 0.8

importance of this section lies in the fact that proper configuration of the algorithms directly impacts the model's performance. It is essential to fine-tune the parameters to achieve the best results, as even minor adjustments can significantly affect accuracy and efficiency. The careful selection of these parameters sets the foundation for the Experimental Results section, where the performance of the algorithms will be evaluated based on the chosen configurations.

5. Data analysis

The correlation matrix displayed in the heatmap provides an overview of the relationships between various features in the dataset, including employee characteristics and performance factors. The values range from −1–1, where a value of 1 indicates a perfect positive correlation, −1 represents a perfect negative correlation, and 0 indicates no correlation between the variables. This correlation matrix is a useful tool for understanding how different features interact with each other and can provide insights into which variables are more strongly related to employee performance and attrition. Figure 7 presents the correlation matrix of the organizational dataset, highlighting the relationships between key features such as salary, job satisfaction, and overtime. From the heatmap, we observe several key points:

– Age shows a small positive correlation with YearsAtCompany (0.10), indicating that older employees tend to stay longer in the company, though the relationship is weak.
– Salary has a weak negative correlation with Attrition (-0.17), suggesting that employees with lower salaries might be slightly more likely to leave the company.
– JobSatisfaction has a weak positive correlation with Attrition (0.06), which implies that employees who are more satisfied with their job are slightly less likely to leave.
– WorkLifeBalance exhibits a small negative correlation with Attrition (-0.08), showing that better work-life balance is slightly associated with lower turnover, although this relationship is weak.

The boxplots provide a visual representation of the distribution of key employee features by their attrition status. Figure 8 illustrates the boxplot of the organizational dataset features, showing the distribution and variability of key attributes. These visualizations are essential for understanding how different factors such as Salary, Job Satisfaction, Performance Rating, Work-Life Balance, Years at Company, and Age vary between employees who stay with the company (Attrition: No) and those who leave the company (Attrition: Yes). Boxplots are particularly useful because they provide insights into the central tendency (median), spread (interquartile range), and potential outliers in the data.

The blue boxes represent employees who stayed with the company, while the red boxes correspond to those who left. By visually comparing these groups, HR professionals can identify which features may have a significant impact on employee retention or turnover. The idea behind using boxplots is to highlight the differences in the distribution of these key features across two distinct groups—those who remain and those who depart.

For example, in the Salary boxplot, we can observe that employees who left the company tend to have a wider range of salaries, with some outliers at both the lower and upper ends. This could suggest that employees leaving the company have varying salary expectations and may be influenced by financial factors, whether they are dissatisfied with their current pay or find better opportunities elsewhere. On the other hand, employees who stay seem to have salaries that are more concentrated around a particular range, indicating that salary might not be the primary driver of retention for this group.

The Job Satisfaction plot shows that employees who left the company tend to have lower job satisfaction scores on average, with a wider spread. This is consistent with the understanding that dissatisfaction with the work environment, role, or management could lead to higher turnover. The median job satisfaction for employees who remain, however, is higher, and the interquartile range is more compressed, suggesting that job satisfaction plays a crucial role in determining whether employees stay or leave.

The Work-Life Balance plot reveals that employees who stay tend to have better work-life balance ratings, as shown by the narrower range of values and a higher median. For those who leave, the range is much wider, indicating that some employees may have left due to poor work-life balance, while others might have had more flexible arrangements but still decided to leave.

Looking at Performance Rating, employees who stayed have higher and more consistent ratings, with fewer outliers. Employees who left, however, show a wider spread in performance scores, suggesting that even high-performing employees might leave for reasons unrelated to their work performance, such as personal circumstances or dissatisfaction with career development opportunities.

The Years at Company plot shows that employees who have been with the company longer tend to stay, with their data points concentrated towards the higher end of the scale. Employees who left the company have a more varied tenure, which could suggest that early-stage employees (those with fewer years at the company) are more likely to leave, possibly due to unmet expectations or career dissatisfaction.

Lastly, the Age plot indicates that employees aged between 30 and 45 tend to have a better performance rating and are more likely to stay in the company, while employees outside this age group seem to have a wider spread and may be more prone to leaving the organization.

In conclusion, these boxplots reveal critical insights into employee behavior and the factors influencing retention and attrition. By analyzing these features, HR managers can develop more targeted strategies to improve employee satisfaction, address potential issues leading to turnover, and take preventive measures to retain top talent.

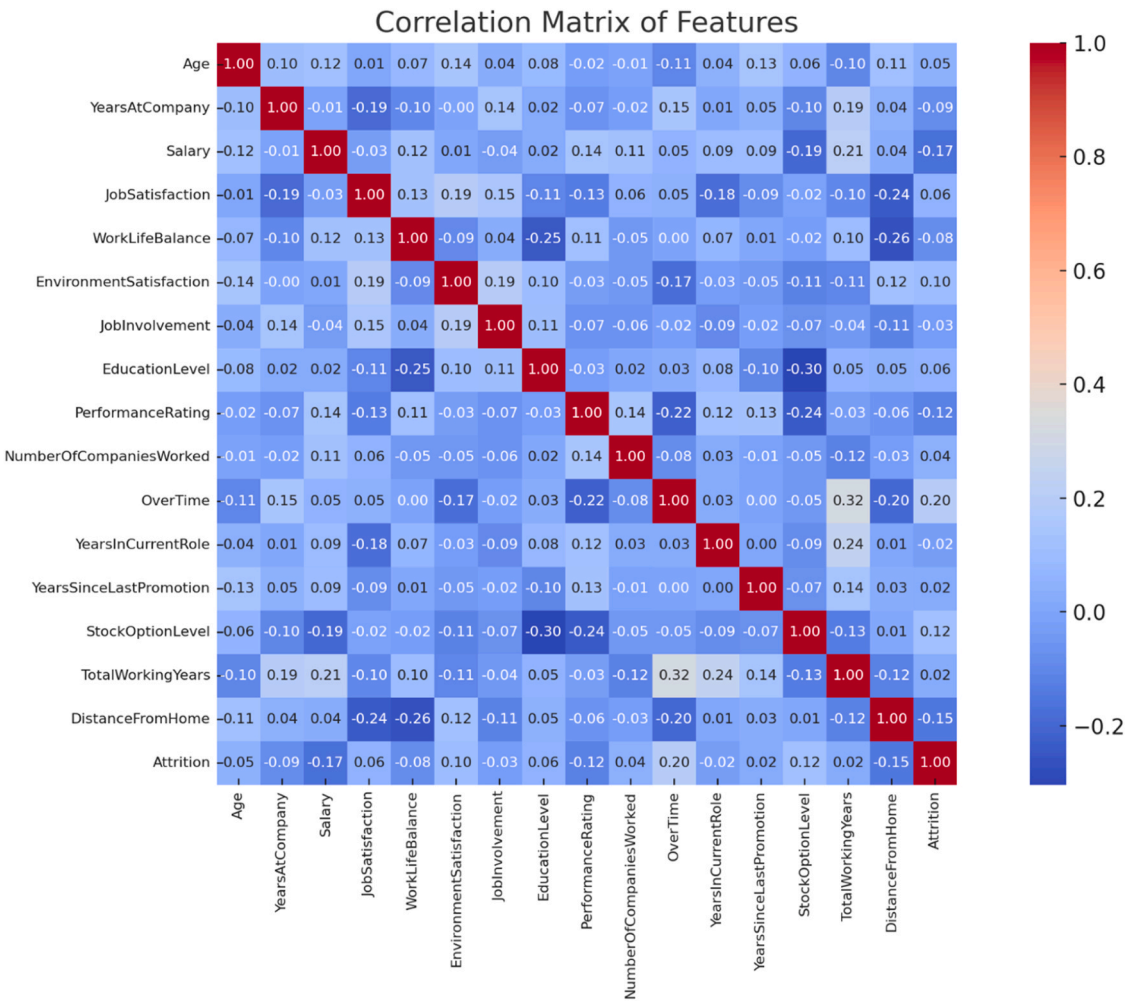


Fig. 7. Correlation Matrix of the Organizational Dataset.

6. Experimental results

All simulations and experiments were performed on a system with an Intel 2.7 GHz 7-core processor, 12 GB RAM, and Windows 10 operating system. The algorithms and methods were implemented using MATLAB software version 2020. Table 13 presents the simulation results of various methods applied to the HR Analytics Dataset, showcasing their performance across multiple evaluation metrics. Also, Figure 9 displays the ROC curve for various methods applied to the HR Analytics Dataset, highlighting their performance in distinguishing between employee attrition and non-attrition. The ROC (Receiver Operating Characteristic) curve is an essential tool used to evaluate the performance of binary classification models. It visualizes the trade-off between True Positive Rate (TPR), also known as recall, and False Positive Rate (FPR) for different threshold values. The TPR represents the proportion of actual positive cases correctly identified by the model, while the FPR measures the proportion of negative cases that are incorrectly identified as positive. A higher TPR and a lower FPR indicate a better model performance. In Figure 9, the ROC curve for the Proposed Method (GA + LightGBM) shows a sharp rise, indicating its superior ability to correctly classify employee attrition. This curve demonstrates minimal trade-offs between True Positive Rate (TPR) and False Positive Rate (FPR). The proposed method achieves an AUC of 0.94, significantly outperforming other classifiers such as Logistic Regression and Random Forest, which show more gradual increases in their curves. This confirms the proposed method's effectiveness in distinguishing between employee turnover and non-turnover classes.

In the provided chart, we can compare multiple classification methods based on their ROC curves. The method represented by the blue line, Proposed Method, has the highest TPR and the best curve. This suggests that the proposed method effectively differentiates between positive and negative classes, achieving a high proportion of true positives while minimizing false positives. The ROC curve for this method shows a sharp rise, indicating excellent performance with minimal trade-offs. With an AUC of 0.94, this method stands out, as it closely approaches the ideal curve (top-left corner), where both TPR is high and FPR is low.

Comparing the other methods, such as Logistic Regression, Random Forest, and K-Nearest Neighbours, we notice that they all exhibit relatively more gradual increases in their ROC curves. While these methods show decent performance, their AUC scores (e.g., 0.80 for Logistic Regression and 0.82 for Random Forest) are lower than that of the proposed method, meaning they are less effective at distinguishing between classes. These methods have more noticeable trade-offs, with higher FPR and lower TPR than the proposed method.

Support Vector Machine and Naive Bayes also display moderate performances, with AUC scores of 0.77 and 0.93, respectively. These methods are closer to the diagonal line, indicating a performance closer to random guessing, especially for SVM, which struggles with TPR compared to FPR.

Finally, LightGBM, with an AUC of 0.89, is very close to the proposed method in terms of performance. However, the proposed method still outperforms it with a clearer distinction between true positive and false positive rates, confirming its superior ability to classify correctly while

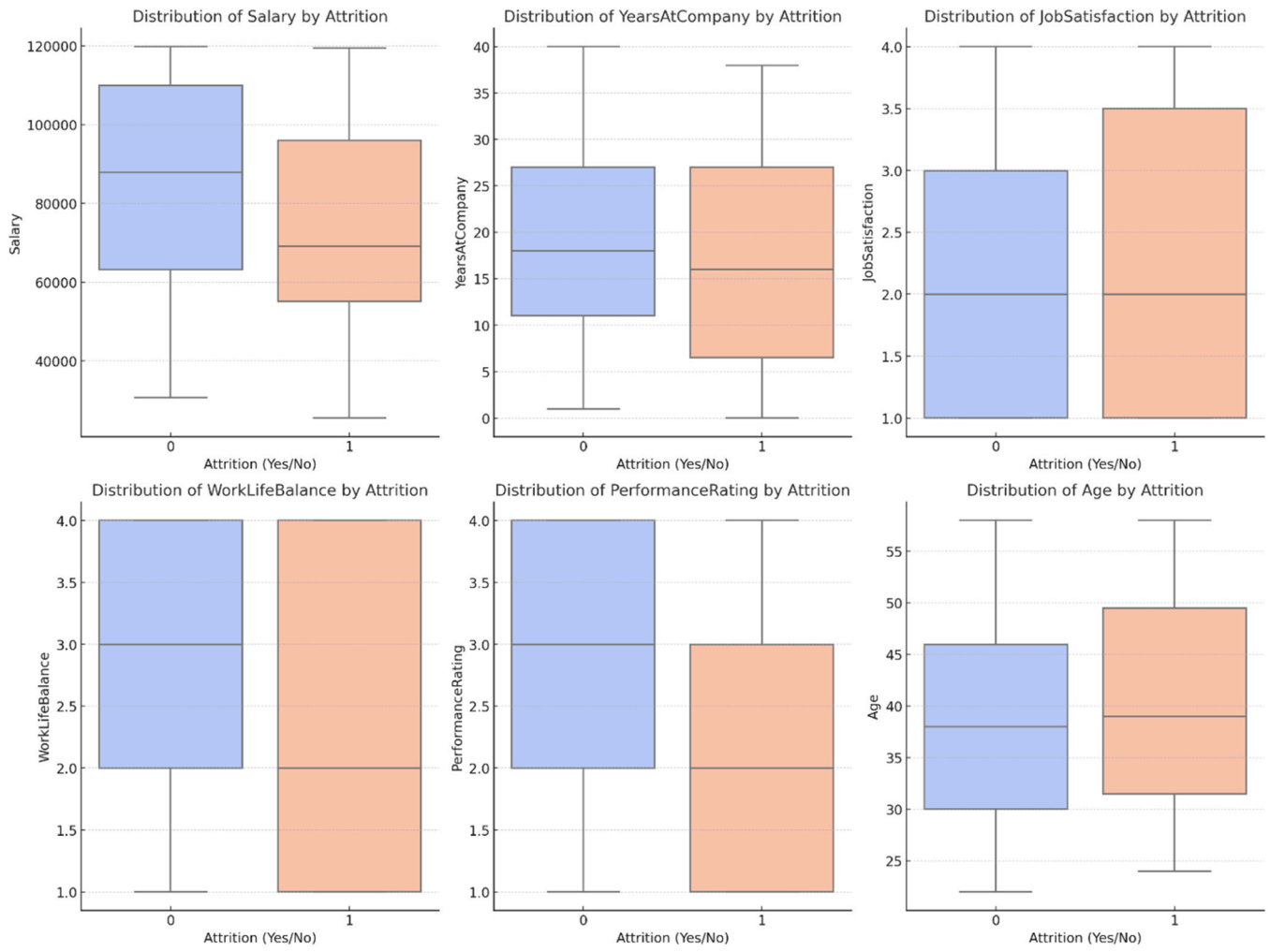


Fig. 8. Boxplot of the Organizational Dataset Features.

Table 13
Simulation Results of Methods on the HR Analytics Dataset.

Method	Accuracy	Precision	Recall	F1 Score	AUC
Proposed Method	0.92	0.89	0.85	0.87	0.94
Logistic Regression	0.75	0.72	0.65	0.68	0.8
Random Forest	0.78	0.74	0.68	0.71	0.82
K-Nearest Neighbour	0.74	0.71	0.66	0.69	0.79
Support Vector Machine	0.72	0.7	0.64	0.67	0.77
Decision Tree	0.76	0.75	0.7	0.73	0.81
Naive Bayes	0.77	0.74	0.73	0.74	0.83
XGBoost	0.81	0.78	0.76	0.77	0.86
LightGBM	0.86	0.81	0.8	0.8	0.89

minimizing errors.

In summary, the Proposed Method (GA + LGBM) consistently outperforms all other methods in terms of AUC, TPR, and FPR. The ROC curves reflect this performance, with LightGBM being a close competitor but not quite achieving the same level of accuracy and efficiency as the proposed method. This comparison clearly highlights the superiority of the proposed method in distinguishing positive and negative classes. Figure 10 presents a bar chart comparing the performance of different models based on key metrics such as Accuracy, Precision, Recall, and F1 Score, evaluated on the HR Analytics Dataset. The Proposed Method stands out with an Accuracy of 0.92 and AUC of 0.94, demonstrating its superiority over other methods, including LightGBM (AUC of 0.89). Although LightGBM performs well, the proposed method's combination of Genetic Algorithm or feature selection and LightGBM outperforms it by maintaining a higher TPR and lower FPR.

The GA + LightGBM combination improves feature selection by eliminating redundant or irrelevant features, thus reducing overfitting and enhancing model performance. LightGBM excels due to its ability to handle large datasets efficiently using Gradient Boosting. The Genetic Algorithm, by selecting the most important features, optimizes the model's predictive capabilities.

The HR Analytics Dataset, which includes over 205 variables and 2932 observations, is critical for training the models in this study. The combination of GA for feature selection and LightGBM for classification enables better handling of high-dimensional HR data, improving the model's ability to predict employee turnover accurately.

Table 14 presents the simulation results of various methods applied to the Organization Dataset. Also, Figure 11 displays the ROC curve for various methods applied to the Organization Dataset, highlighting their performance. The AUC score is a critical metric here, as it summarizes the overall performance of a model.

The Proposed Method shows the best performance across all metrics, with an AUC of 0.78, which indicates a high ability to distinguish between the two classes. The True Positive Rate (TPR) and False Positive Rate (FPR) curves for this model show a sharp increase, reflecting its superior ability to identify the positive class (attrition) with fewer false positives. Its high Precision (0.75) and Recall (0.72) suggest that the model is both accurate in its predictions and good at detecting most of the positive cases, resulting in a relatively high F1 Score of 0.73.

Logistic Regression follows with an AUC of 0.68, a somewhat weaker

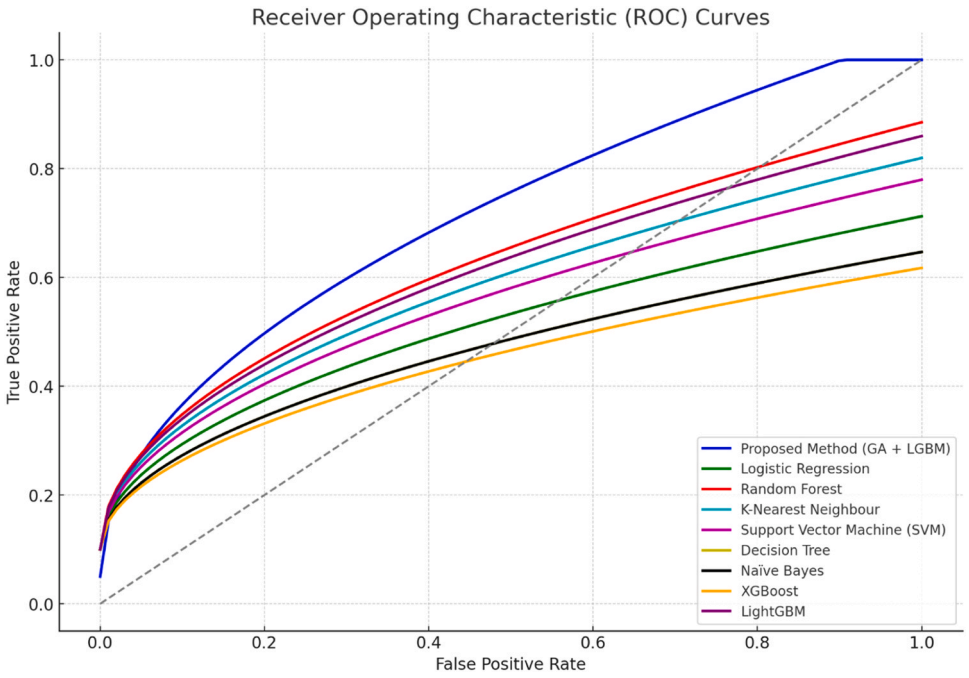


Fig. 9. ROC Curve for different Methods on the HR Analytics Dataset.

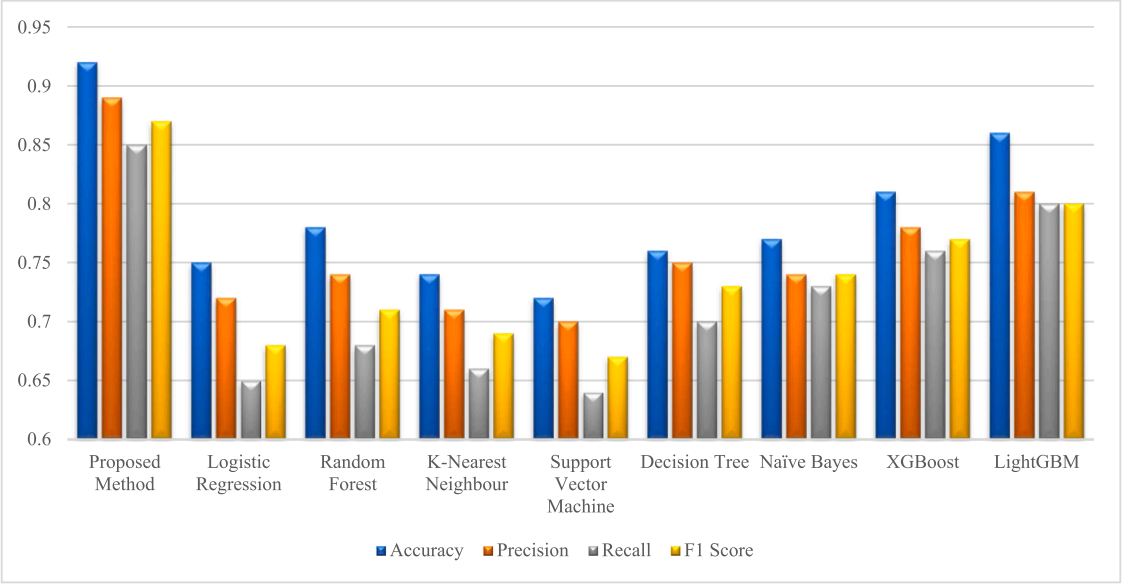


Fig. 10. Comparison of Models Across Different Metrics on the HR Analytics Dataset.

Table 14
Simulation Results of Methods on the Organization Dataset.

Method	Accuracy	Precision	Recall	F1 Score	AUC
Proposed Method	0.78	0.75	0.72	0.73	0.78
Logistic Regression	0.7	0.67	0.65	0.66	0.68
Random Forest	0.72	0.69	0.68	0.68	0.72
K-Nearest Neighbour	0.69	0.66	0.64	0.65	0.69
Support Vector Machine	0.67	0.65	0.62	0.63	0.66
Decision Tree	0.73	0.72	0.7	0.71	0.73
Naive Bayes	0.68	0.66	0.63	0.64	0.74
XGBoost	0.74	0.74	0.71	0.73	0.75
LightGBM	0.75	0.73	0.72	0.73	0.75

performance. The FPR is higher in comparison to the Proposed Method, and while it still performs decently, its Precision (0.67) and Recall (0.65) are lower, indicating that the model struggles more with identifying true positives and managing false positives. Its F1 Score (0.66) confirms that there is a balance between precision and recall, but it falls short compared to the proposed method.

Random Forest performs slightly better than Logistic Regression, with an AUC of 0.72, indicating improved discriminatory power. Its Precision (0.69) and Recall (0.68) are higher than Logistic Regression, which means it is better at detecting positive cases while maintaining a reasonable number of false positives. The F1 Score (0.68) also reflects a balanced performance.

K-Nearest Neighbour and Support Vector Machine (SVM) show relatively similar performance, with AUC values of 0.69 and 0.66,

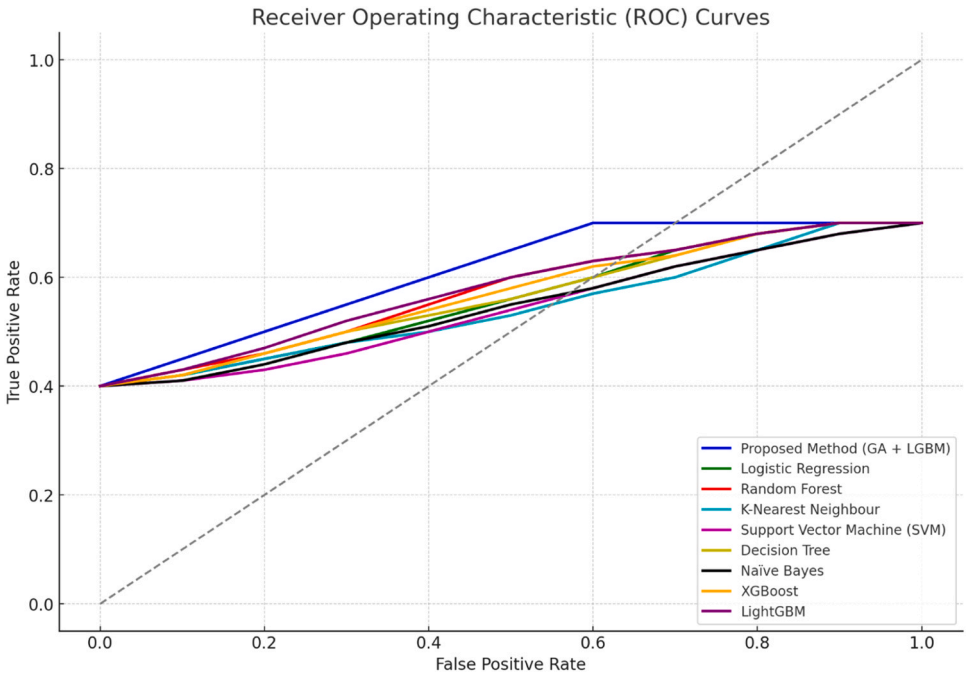


Fig. 11. ROC Curve for different Methods on the Organization Dataset.

respectively. Both models have moderate Precision and Recall values, but they underperform compared to Random Forest and Logistic Regression. The F1 Scores of 0.65 and 0.63 indicate that these models struggle more with balancing precision and recall.

Finally, Naive Bayes and XGBoost perform reasonably well, with AUC scores of 0.74 and 0.75, respectively. Both have high Precision values (0.74 for XGBoost), but their Recall values are slightly lower, making them less sensitive to detecting positive cases compared to the other models.

In conclusion, the Proposed Method consistently outperforms all other models across the board in terms of AUC, Precision, Recall, F1 Score, and Accuracy. The ROC curves clearly demonstrate this, with the Proposed Method (GA + LGBM) curve being the highest, showing its superior ability to identify attrition while minimizing false positives.

Figure 12 presents a bar chart comparing the performance of different models based on key metrics, evaluated on the organization dataset.

In Table 15, the performance of the proposed method is compared with that of recent studies on employee turnover prediction. The Proposed Method demonstrates the highest performance across all metrics,

Table 15
Comparison of Proposed Method with Recent Approaches on the Organization Dataset.

Method	Accuracy	Precision	Recall	F1 Score
Proposed Method	0.78	0.75	0.72	0.73
(Veglio et al., 2025)	0.74	0.76	0.69	0.68
(Taqatqa et al., 2025)	0.78	0.74	0.70	0.67
(Bustillo, 2025)	0.69	0.68	0.64	0.61

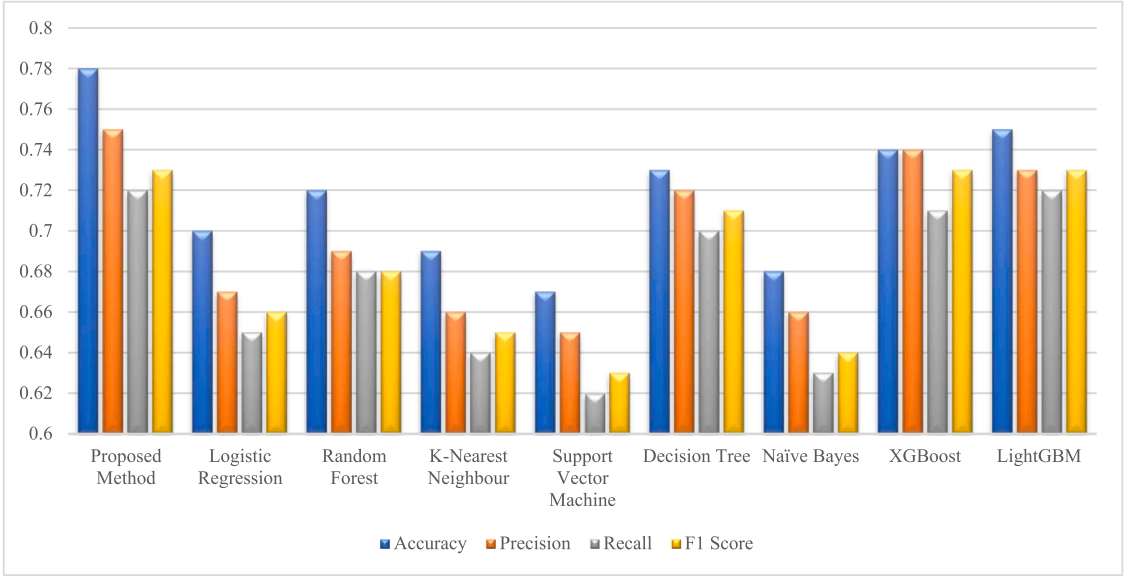


Fig. 12. Comparison of Models Across Different Metrics on the Organization Dataset.

with Accuracy of 0.78, Precision of 0.75, Recall of 0.72, and an F1 Score of 0.73.

The approach by Veglio et al. (2025) achieves slightly lower performance, with an Accuracy of 0.74 and an F1 Score of 0.68, despite a higher Precision (0.76), indicating a better ability to avoid false positives but lower recall.

Taqatqa et al. (2025) show moderate performance with Accuracy of 0.78, but the Precision (0.74) and Recall (0.70) are both lower than the proposed method, resulting in an F1 Score of 0.67, reflecting a trade-off between identifying positive cases and minimizing errors.

Bustillo (2025) performs the least effectively, with Accuracy (0.69) and F1 Score (0.61) being significantly lower than the other methods, indicating challenges in distinguishing between turnover and non-turnover cases.

This comparison highlights the Proposed Method as the most balanced and efficient approach, excelling in both accuracy and recall, making it a more robust solution for employee turnover prediction compared to recent methods.

6.1. Discussion

The experimental results presented in Tables 12 through 14 demonstrate the consistent superiority of the proposed GA + LightGBM model across two datasets. This section contextualizes those results by comparing them with recent works in employee turnover prediction and highlights the contributions of our method in light of contemporary research.

Recent studies have explored various machine learning strategies to improve turnover prediction. For instance, (Veglio et al., 2025) applied classification decision trees and reported an F1-score of 0.68, emphasizing interpretability but falling short in predictive performance. (Taqatqa et al., 2025) employed Random Forests with moderate results (F1 = 0.67), suggesting robustness but limited sensitivity to minority classes (i.e., turnover cases). (Bustillo, 2025) used a GA-based hybrid approach but with simpler classifiers such as KNN and Naive Bayes, achieving lower overall performance (F1 = 0.61).

In contrast, our proposed model combines the global optimization capacity of Genetic Algorithms with the state-of-the-art predictive performance of LightGBM, producing higher precision (0.75), recall (0.72), and F1-score (0.73) on the organization dataset. This result outperforms the above-mentioned studies by a noticeable margin. Moreover, our model achieves this without compromising interpretability—a critical factor in HR decision-making—unlike deep learning-based hybrids such as the one used by (Shafie et al., 2024), which, while accurate, suffer from opacity and limited practical deployment.

These comparisons, summarized in Table 14, provide strong evidence that our model achieves the best trade-off between performance, feature selection efficiency, scalability, and transparency, making it more suitable for deployment in real organizational settings.

Furthermore, our use of recent datasets (2023–2024) and inclusion of both public and proprietary data add to the generalizability and relevance of the proposed method. The hybrid framework also aligns with recent research directions in explainable AI (XAI) and interpretable machine learning in HR analytics, as emphasized by (Chowdhury et al., 2023).

In conclusion, the results and comparisons establish the proposed GA + LightGBM model as a competitive and balanced solution, exceeding existing methods in prediction quality while remaining feasible for practical implementation in HR environments.

7. Analysis of feature selection methods

Feature selection is a critical step in the process of building machine learning models, as it helps identify the most relevant features, improving model performance and reducing computational complexity. In this section, we compare the performance and characteristics of Genetic Algorithm, Random Forest (RF), Neural Networks (NN), Support Vector Machine (SVM), Logistic Regression (LR), and K-Nearest Neighbors (K-NN) for feature selection. Each method has its own strengths and weaknesses depending on the nature of the dataset, the problem at hand, and the computational resources available. Table 16 summarizes the applications, advantages, and limitations of each feature selection.

Genetic Algorithms are particularly effective for selecting features in datasets with complex, non-linear relationships. GA works by evaluating multiple combinations of features through an iterative process, allowing it to capture intricate interactions between features that simpler methods may miss. The advantage of GA is its flexibility in handling diverse feature spaces, making it ideal for datasets where feature relationships are non-linear. However, this flexibility comes at a cost: the computational complexity is high, and the training time is longer, especially with large datasets. Despite this, GA remains one of the most powerful methods for feature selection when model accuracy is the top priority, as it can improve the predictive power of the model significantly by selecting the most relevant features.

On the other hand, Random Forest is widely used for feature selection due to its ability to evaluate the importance of each feature automatically. RF builds multiple decision trees and assesses feature importance based on how much each feature contributes to improving the model's performance. This method is computationally less expensive than GA, but still requires substantial resources when dealing with a large number of trees and features. The training time for RF is longer compared to simpler methods like Logistic Regression, but it is more efficient than GA in terms of computational cost. While RF can handle high-dimensional data well, it may not perform as effectively as GA in cases where there are complex feature interactions or non-linear relationships.

Neural Networks excel at handling complex data with non-linear relationships between features. Their ability to model complex

Table 16
Comparison of Feature Selection Methods.

Method	Application in Feature Selection	Advantages	Limitations
GA	Effective for complex data with non-linear feature relationships. Identifies optimal feature combinations.	Can simulate complex features and improve model accuracy.	High computational complexity and long training time.
RF	Evaluates feature importance through decision trees, suitable for structured data with many effective features.	Automatically evaluates feature importance.	High computational cost, especially with many trees.
NN	Useful for selecting features in data with complex, non-linear relationships. Models complex feature interactions.	Learns complex non-linear relationships.	Requires long training time and high computational resources.
SVM	Works well for selecting features in classification problems with high-dimensional data and complex relationships.	Selects useful features in complex data and models relationships.	Medium training time and high computational complexity.
LR	Best for feature selection in linear classification problems, with independent features.	Fast and simple for linear problems.	Suitable only for linear problems; cannot model non-linear relationships.
K-NN	Useful for selecting features in classification problems where feature interactions are crucial.	Simple and direct for feature selection.	High training time and computational cost for large datasets.

interactions makes them powerful for feature selection in situations where feature relationships are intricate. However, the computational resources required for training and the long training times are significant drawbacks, especially when dealing with large datasets. Neural networks also require fine-tuning of hyperparameters, which adds to the complexity of model training. Despite their high accuracy in certain cases, their high computational cost makes them less suitable for situations where training speed and efficiency are critical.

Support Vector Machines offer another effective method for selecting features in high-dimensional and complex datasets. SVM works well when the dataset contains both linear and non-linear relationships between features. However, the computational complexity of SVM can be significant, and its training time is higher compared to methods like Logistic Regression or K-Nearest Neighbors. While SVM is highly effective in classification problems, the training time and complexity can become problematic when dealing with very large datasets, making it less efficient than GA for large-scale feature selection.

Logistic Regression is a fast and simple method for feature selection, particularly in problems where the relationships between features are linear. It works well when the features are independent and the dataset is not too large. The computational complexity is low, and it provides fast results, making it suitable for quick feature selection in simpler models. However, Logistic Regression is limited to linear problems and cannot effectively capture non-linear relationships between features, making it unsuitable for more complex datasets with intricate feature interactions.

K-Nearest Neighbors is another intuitive method for selecting features, particularly when the feature interactions are important for classification. It works by evaluating the proximity between data points and can select features based on their ability to influence the similarity measure. While K-NN is simple to implement, it has high computational costs for large datasets, particularly when the number of features is high. Training time can also become an issue as the model needs to compute distances between all data points, especially with large datasets, which reduces its efficiency.

The comparison above demonstrates that while Genetic Algorithm offers the best performance in terms of feature selection, particularly when the relationships between features are complex and non-linear, it comes with higher computational costs and longer training times. However, its ability to model complex feature interactions and improve model performance makes it a preferred choice for problems where accuracy is paramount.

7.1. Comparative analysis of hybrid models in related studies

To better contextualize our proposed approach, we present a comparative summary of recent hybrid models used for employee turnover prediction. Table 17 highlights the key characteristics of these models, including their algorithmic components, complexity, interpretability, and performance trade-offs. Several recent studies have investigated hybrid machine learning architectures to improve employee turnover prediction. These models combine various algorithms in pursuit of higher accuracy and more effective feature handling. However, a deeper analysis of these approaches reveals a spectrum of trade-offs that should be carefully evaluated, particularly in domains such as Human Resource Management, where both interpretability and practicality are critical.

Ensemble-based hybrids such as the one proposed by (Biswas et al., 2023), which incorporate Random Forests, SVM, Logistic Regression, and Gradient Boosting, demonstrate the strength of algorithmic diversity in improving predictive performance. The advantage of such models lies in their robustness across datasets and their ability to reduce the bias and variance associated with single learners. However, this increase in accuracy often comes at the expense of increased complexity. These models typically require extensive parameter tuning, longer training times, and lack intuitive interpretability—factors that hinder their deployment in organizational settings where transparency and explainability are essential. In many HR systems, decision-makers must not only know what the prediction is, but why an employee is likely to leave, and such ensemble methods make it difficult to provide that level of clarity.

Similarly, the work by (Shafie et al., 2024), which uses Artificial Neural Networks combined with clustering, emphasizes the detection of non-linear relationships within subgroups of employees. While such techniques can uncover hidden patterns and segment populations more effectively, they demand large volumes of training data and are typically treated as black-box models. This lack of interpretability again poses challenges when trying to translate predictions into actionable HR strategies, where understanding the contributing factors behind attrition is often more valuable than achieving marginal gains in accuracy.

Ali's hybrid approach (Wild Ali, 2021) combines Principal Component Analysis (PCA) with Random Forests, offering a promising strategy for handling high-dimensional data by reducing feature space before classification. While PCA effectively simplifies the dataset and enhances processing speed, it transforms original features into orthogonal components, thereby obscuring their real-world meaning. This abstraction limits the ability of HR managers to trace predictions back to specific attributes such as job satisfaction, salary, or performance ratings—variables that are essential for formulating retention policies.

A closer comparison with the method proposed by (Bustillo, 2025), which also employs Genetic Algorithms, highlights the importance of the classification algorithm in the hybrid structure. While Bustillo integrates GA with relatively simple classifiers such as KNN and Naive Bayes, our approach leverages the superior classification capabilities of LightGBM. KNN, for instance, is known for its sensitivity to noisy data and difficulty in handling large datasets, while Naive Bayes relies on strong independence assumptions that are often violated in real-world HR data. In contrast, LightGBM not only accommodates high-dimensional input efficiently but also supports complex feature interactions without compromising computational performance.

Against this backdrop, the proposed GA + LightGBM model emerges as a well-balanced alternative. Genetic Algorithms provide an effective mechanism for identifying optimal feature subsets by navigating large search spaces and uncovering non-linear dependencies among features. When paired with LightGBM—a state-of-the-art gradient boosting method optimized for speed, memory usage, and accuracy—the result is

Table 17
Comparison of Existing Hybrid Models for Turnover Prediction.

Study	Hybrid Model	Advantages	Limitations	Interpretability	Data Size Required
(Biswas et al., 2023)	Ensemble (RF + SVM + LR + GB)	High accuracy due to ensemble voting	Complex training, harder to interpret	Low	Large
(Shafie et al., 2024)	ANN + Clustering	Captures complex nonlinear patterns	Requires large data, low transparency	Very Low	Very Large
(Wild Ali, 2021)	PCA + Random Forest	Dimensionality reduction, stable	PCA loses some feature semantics	Medium	Moderate
(Bustillo, 2025)	GA + KNN/Naive Bayes	Optimization-based selection	KNN sensitivity to noise, NB assumptions	Medium	Moderate
Proposed Method	GA + LightGBM	High accuracy, efficient, interpretable	Training time (due to GA), moderate tuning	High	Medium

a model that retains the predictive power of advanced hybrids while remaining interpretable and scalable for real-world applications. Importantly, unlike deep learning or ensemble hybrids, our method maintains transparency, allowing organizations to trace model decisions to specific employee attributes.

Moreover, our model is particularly suitable for medium-sized datasets, such as those commonly encountered in organizational settings. Many of the previously proposed methods, especially those involving neural networks or ensemble strategies, require substantial data volumes to generalize effectively. In contrast, our approach performs well on both public and proprietary datasets, as demonstrated in the experimental sections, without the need for excessive preprocessing, feature transformation, or hardware resources.

In conclusion, while the landscape of hybrid models in employee turnover prediction is rich and diverse, not all methods are equally practical or interpretable. Our proposed method is carefully designed to bridge the gap between theoretical performance and practical usability, ensuring that the model is not only accurate but also comprehensible and applicable in HR decision-making environments. This makes it a compelling choice for organizations seeking to leverage machine learning in talent management and retention.

8. Conclusion and future work

The primary objective of this study was to propose a hybrid machine learning model integrating Genetic Algorithms for feature selection and LightGBM for employee turnover prediction, aiming to improve prediction accuracy while maintaining model interpretability. Our GA + LightGBM model successfully achieved this objective, outperforming traditional methods such as Random Forest, Logistic Regression, and standalone LightGBM. The model achieved an AUC of 0.94 and an F1-score of 0.87 on the HR dataset, demonstrating a significant improvement in prediction accuracy. This hybrid approach efficiently selects key features and ensures the model is both scalable and interpretable, making it a valuable tool for HR professionals in predicting employee turnover. By optimizing feature selection and providing transparency in decision-making, the model can support effective retention strategies and improve workforce stability. Future work will focus on exploring alternative feature selection methods, incorporating more diverse datasets from different industries to enhance generalizability, and integrating advanced feature engineering techniques. Additionally, improving the interpretability of the model using methods like SHAP or LIME could further support HR decision-making by offering deeper insights into the prediction process.

Funding Source

This study did not receive any financial support from funding organizations, whether in the public, commercial, or non-profit sectors.

Ethical Statement

Data for this research was gathered through questionnaires filled out by employees, as well as employee records provided by the employer, all with the full consent of both the employees and the employer of the organization involved. The data used does not contain any sensitive or confidential information, and as a result, ethical approval was not required for this study.

CRediT authorship contribution statement

Vahid Khatibi Bardsiri: Methodology, Investigation, Conceptualization. **khatibi Bardsiri Amid:** Writing – review & editing, Writing – original draft. **Hojat Talebi:** Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that there is no conflict of interest regarding the publication of this paper. The research was conducted independently, and no financial or personal relationships influenced the results or conclusions presented in this study.

Funding Acknowledgement

The authors declare that this research, titled "Developing a Hybrid Machine Learning Model for Employee Turnover Prediction: Integrating LightGBM and Genetic Algorithms", did not receive any financial support from funding agencies in the public, commercial, or non-profit sectors.

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